

# Do ice and subglacial observables determine basal friction?

## A point-wise test in the Amundsen Sea sector

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**ABSTRACT.** Uncertainties in basal-sliding parameterizations hinder accurate projections of ice-sheet contributions to sea-level rise. Although inverse methods can calibrate basal friction to reproduce observed velocities, the controls on inversion-derived friction remain poorly understood. We test whether ice and subglacial observables can define a consistent point-wise relationship with the inversion-derived basal-friction control across the Amundsen Sea sector of West Antarctica. Using a sector-wide inversion for the basal-friction control as the reference, we train neural-network regression models in ten experiments, each with a spatially separated 50 km square and buffer withheld from training and validation. We test each friction-control field in a forward simulation against observed velocity. Depending on predictor configuration, the friction fields reduce modeled-velocity error relative to a spatially uniform friction baseline in six to nine of the ten squares. Performance varies among locations, adding predictors does not consistently improve performance, and failures occur even where marginal and joint predictor support is high. Although the tested local predictors improve velocity in most withheld squares, modeled-velocity errors remain substantially larger than for the inversion reference. These results indicate that the considered ice and subglacial observables are insufficient to define a consistently successful point-wise parameterization of inversion-derived basal friction.

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## INTRODUCTION

Ice-sheet models are used to project sea-level rise, reconstruct past ice-sheet behavior, and investigate the physical processes governing ice-mass change (e.g. Lecavalier and Tarasov, 2025; Nowicki and others, 2016). These models contain structural uncertainty arising from incomplete representations of physical processes and parameter uncertainty arising from imperfectly constrained model parameters (Hebeler and others, 2008; Recinos and others, 2023). Both sources of uncertainty limit our ability to constrain probabilistic projections, especially for low-probability, high-impact outcomes (Kopp and others, 2017).

Basal slip is an important source of both structural and parameter uncertainty (Joughin and others, 2019; Bulthuis and others, 2019). Motion at the ice–bed interface can involve sliding over hard bedrock, deformation of subglacial sediment, and changes in basal water pressure, among other processes (Schoof and Hewitt, 2013; Cuffey and Paterson, 2010). Ice-sheet models generally represent these processes through a sliding law containing one or more spatially varying friction parameters (Weertman, 1964; Schoof, 2005). Inverse methods adjust these parameters to reduce the mismatch between modeled and observed quantities such as velocity (MacAyeal, 1993). The resulting friction field therefore absorbs unresolved basal processes, including bed roughness, till rheology, and hydrology (Hoffman and others, 2022; Tulaczyk and others, 2000; Lu and Kingslake, 2024), as well as geometry errors and differences among model formulations (Zhao and others, 2018; Barnes and others, 2021). It is consequently an effective model parameter rather than a direct observation of basal conditions.

Inversion studies show that basal resistance across Antarctica is spatially heterogeneous (Morlighem and others, 2013; Zhao and others, 2018). This motivates a working hypothesis: at the resolution of an ice-sheet model, the dimensionless basal-friction control  $C(\mathbf{x})$  may be approximated by a point-wise function of observables evaluated at the same location,

$$C(\mathbf{x}) \approx f(\mathbf{z}(\mathbf{x})), \quad (1)$$

where  $\mathbf{z}(\mathbf{x})$  contains locally measured or inferred quantities that may influence basal resistance or serve as proxies for unresolved basal conditions. Equation 1 is a testable point-wise approximation at the model-grid scale, not a physical sliding law. We test it by fitting the relationship outside a held-out region and asking whether the predicted  $C$  improves modeled velocity within that region, particularly where the region’s predictor values resemble those in the data used for fitting.

We divide the candidate predictors into two groups. The ice-product group describes ice geometry, its gradients, driving stress, and surface air temperature. The subglacial-product group describes bed geometry, bed-surface gradient alignment, geothermal heat flux, and gravity and magnetic anomalies. Ice thickness controls overburden pressure and normal stress at the bed, while surface slope governs driving stress; together, variations in thickness and surface elevation gradient impose systematic constraints on basal drag (MacAyeal, 1989; Cuffey and Paterson, 2010, Sections 7.2–7.4 and 8.2). Surface air temperature provides the upper thermal boundary condition and influences the ice sheet’s englacial and basal thermal state (Meierbachtol and others, 2015).

Subglacial conditions can affect basal drag through bed topography and substrate composition. Bed gradients characterize topographic variation that can contribute to form drag (Hoffman and others, 2022), while deformable till can reduce resistance and enable sliding (Cuffey and Paterson, 2010; Tulaczyk and others, 2000; Boulton and others, 2001). Gravity disturbances reflect subsurface density variations associated with bedrock and sedimentary structure (Hackney and Featherstone, 2003). Gravity data can therefore constrain broad bed geometry where direct measurements are sparse (An and others, 2017). Bedrock lithology can influence sediment characteristics, while deformable sediment can reduce basal resistance (Crompton and Flowers, 2016; Boulton and others, 2001). Gravity measurements cannot resolve the thin (centimeter- to meter-scale) sediment layers that affect sliding, but larger density contrasts can identify sedimentary basins; such sediment bodies can influence ice-stream onset (Bell and others, 1998; Studinger and others, 2001; Peters and others, 2006). Magnetic anomalies reflect contrasts in crustal magnetization (Golynsky and others, 2018). Together with gravity and radar data, they can help infer subglacial geology (Jordan and others, 2023) and regional thermal structure (Martos and others, 2017; Reading and others, 2022), providing indirect context for basal drag. Geothermal heat flux supplies heat at the bed and can influence whether basal ice reaches the pressure-melting point. Where basal ice is at the melting point, additional heat can support basal meltwater production and thereby influence basal resistance (Dawson and others, 2022; Kang and others, 2022).

Previous studies have compared inferred basal resistance with radar, seismic, and other observations of the subglacial environment, but have not established whether it follows a consistent local relationship with these observations. Haris and others (2024) found that regions with distinct radar signatures differ in mean basal shear stress, but they compared distributions across regions rather than testing a relationship at individual locations. Das and others (2023) found no robust correlation between inferred sliding parameters

and radar-derived basal reflectivity over Thwaites Glacier, while Kyrke-Smith and others (2017) found no significant kilometer-scale correlation between basal drag and seismic acoustic impedance along Pine Island Glacier (PIG), although values averaged along each seismic profile correlated more strongly.

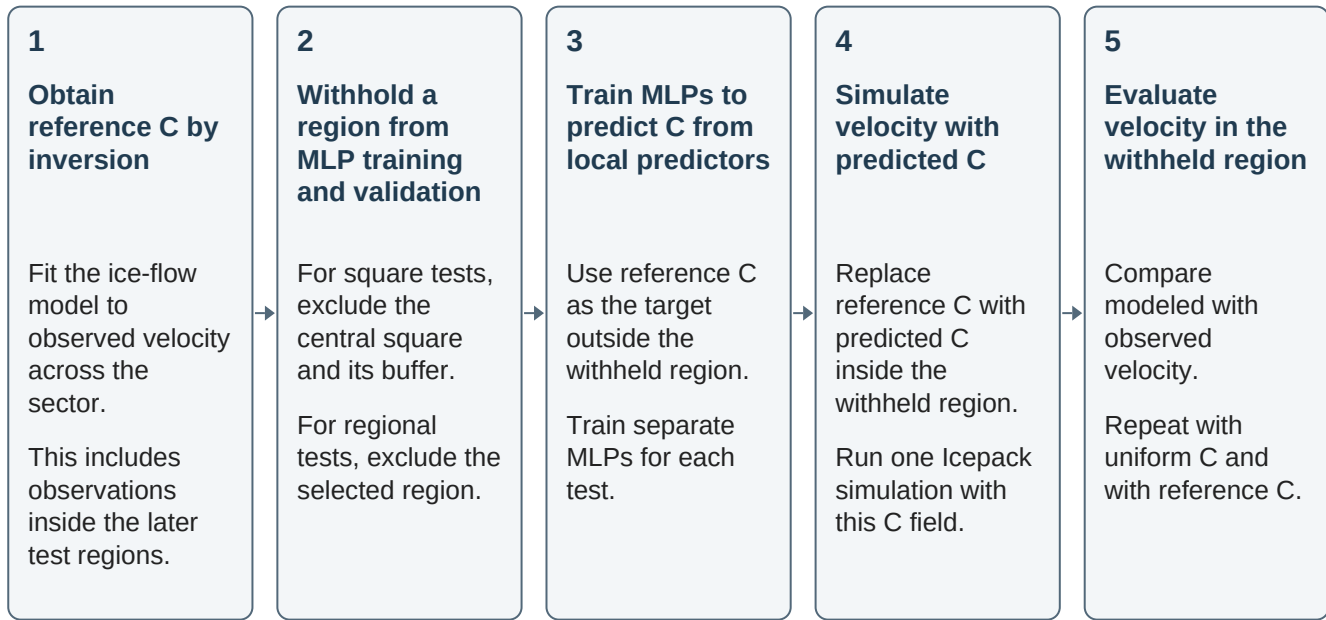
These studies, together with our understanding of basal processes, suggest that the relationship between basal conditions and basal resistance may be complex and nonlinear, involving thermo-mechanical feedbacks, subglacial hydrology, and spatially variable bed properties. Following the field inversion and machine learning approach of Parish and Duraisamy (2016), we use nonlinear regression with a multi-layer perceptron (MLP) as a flexible representation of  $f$  in Eqn 1 to relate the selected observables to the inversion-derived friction control  $C$ .

The point-wise architecture asks whether the selected observables at a location contain enough information to determine the inversion control there, without prescribing a particular functional form. We deliberately exclude observed velocity from the predictor set because changing  $C$  changes the modeled velocity; including velocity as an input would introduce a self-consistency problem (in preliminary tests, iterating  $C$  and velocity to agreement did not converge) and would make the inferred relationship depend directly on the velocity used to evaluate it.

Recent machine-learning studies in glaciology have addressed related but distinct inverse and constitutive problems. Jouvét (2023) inverted a convolutional emulator of Stokes flow, while Jouvét and Cordonnier (2023) and Cheng and others (2024) incorporated ice-flow physics into learned forward or inverse models. Wang and others (2025) used remote sensing and physics-informed deep learning to infer spatially varying ice-shelf rheology. These approaches use spatial structure or governing equations to reproduce ice-flow fields or solve inverse problems. In contrast, the present study uses machine learning only as a flexible function approximator for  $f(\cdot)$  in Eqn 1 to address a different question: do commonly available ice and subglacial observables contain enough information to define a point-wise relationship with inversion-derived basal friction that holds across locations?

## METHODS

This hypothesis is tested in the Amundsen Sea sector as outlined in Fig. 1. A sector-wide basal-friction inversion, fitted to observed velocity over the complete sector, provides the reference  $C$ . We withhold one region from MLP training and validation and evaluate the predicted  $C$  there. We repeat this for eleven withheld regions: ten squares and one catchment; a further test withholding the inter-catchment

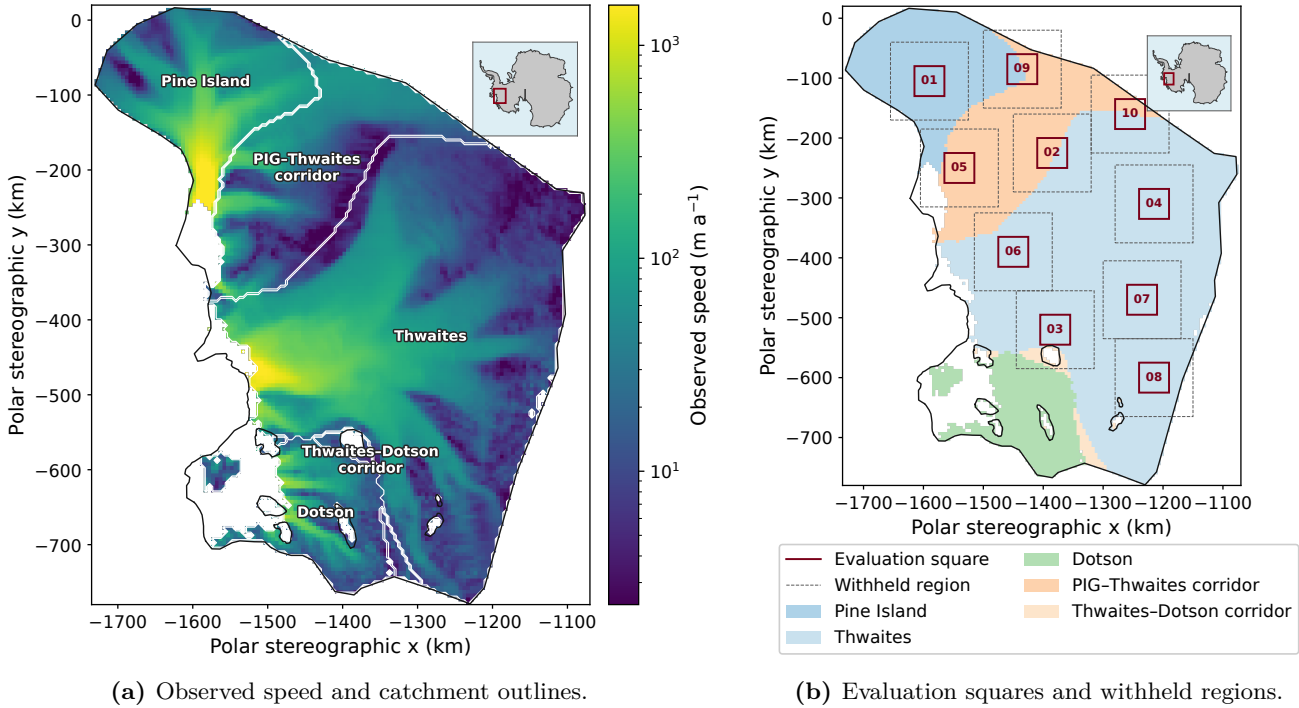


**Fig. 1.** Testing methodology.

corridors is reported in the appendix. In the ten square tests, the withheld region is a 50 km square together with a surrounding 40 km buffer, and the ten squares are spatially separated. In a secondary catchment-scale test, the withheld region is PIG. For each test, separate MLPs are trained to predict the reference  $C$  from local predictors outside that test's withheld region; in the square tests, the other nine squares remain in the training data. The predicted  $C$  replaces the reference  $C$  within the withheld region in an Icepack simulation. We compare the resulting modeled velocity with observed velocity and with two other simulations: one using a spatially uniform  $C$ , as the baseline, and one using the reference  $C$ , as the best case. These tests ask whether a point-wise relationship fitted elsewhere in the sector also holds in the withheld region. We compare six predictor configurations in the square tests. Inversion-derived  $C$  is not necessarily unique (MacAyeal, 1993), so we use modeled velocity as the primary evaluation and treat direct agreement with the reference  $C$  as secondary (Appendix section Basal-friction control: distributions and comparisons).

## Study region and observations

The study domain covers the Pine Island, Thwaites, and Dotson–Crosson drainage systems in the Amundsen Sea sector of West Antarctica (Fig. 2). The catchment outlines were delineated manually, with lateral boundaries following glacier velocity flowlines. PIG, Thwaites, and Dotson are geographic labels; they do not imply that each catchment contains one basal regime.



**Fig. 2.** Study region and holdout design. Panel (a) shows observed MEaSUREs speed over the Amundsen Sea sector, with the PIG, Thwaites, and Dotson regions and the two inter-catchment corridors. Panel (b) shows the ten 50 km primary evaluation squares (solid) and their 130 km withheld region around each (dashed), which adds a 40 km buffer on every side of the square and is excluded from MLP training and validation. Blank areas inside the domain are mountains that are not part of the glacier drainage system and are not modeled.

We take the observed velocity from the Making Earth System Data Records for Use in Research Environments (MEaSUREs) Phase-Based Antarctica Ice Velocity Map, derived from satellite radar interferometry (Mouginot and others, 2019). Surface elevation, ice thickness, and bed elevation are taken from BedMachine Antarctica, version 2 (Morlighem, 2020; Morlighem and others, 2020). BedMachine combines direct observations with reconstruction that uses ice velocity in some regions. The ice-product predictors, and the bed-derived subglacial-product predictors, therefore represent available geometry products, some of which already incorporate velocity information, although velocity itself is not used as a predictor. BedMachine uses different methods to reconstruct ice thickness between observations. Most of our test regions (detailed in the following sections) lie in regions mapped using streamline-diffusion interpolation, which favors smoother thickness variations along flow than across flow (Morlighem and others, 2020).

Surface air temperature and geothermal heat flux follow the products distributed with Quantarctica (Matsuoka and others, 2018; Comiso, 2000; An and others, 2015). The magnetic anomaly is taken from the 5 km EPSG:3031 ADMAP-2S grid (Eagles and others, 2024). The gravity disturbance is taken from the oriented EPSG:3031 AntGG2021 grid (Scheinert and others, 2024).

## Ice-flow model and basal-friction inversion

All simulations used Icepack (Shapiro and others, 2021) and a depth-integrated stress balance based on the shallow-shelf approximation (MacAyeal, 1989). The unstructured triangular mesh has 9,069 vertices and 17,653 elements over 310,546 km<sup>2</sup>, with a median element edge length of 6.6 km. The basal-friction control uses continuous quadratic elements on this mesh, giving 35,797 control points. The shallow-shelf approximation neglects vertical shear, so the modeled horizontal velocity is compared directly with the observed surface velocity. The model parameter  $C_b$  is model-dependent rather than a unique representation of basal conditions as the inferred basal-friction control depends on the adopted stress-balance approximation and basal law.

Basal friction is represented with a Weertman-type law,

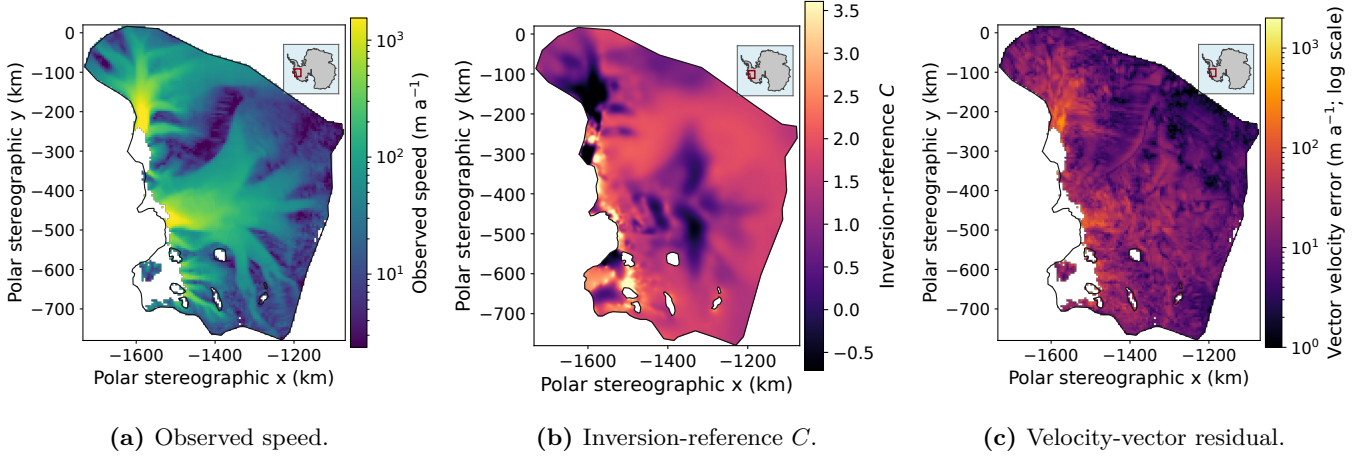
$$\boldsymbol{\tau}_b = -C_b \|\mathbf{u}\|^{1/m-1} \mathbf{u}, \quad (2)$$

where  $\boldsymbol{\tau}_b$  is the basal shear stress on the ice,  $\mathbf{u}$  is depth-averaged horizontal velocity,  $m = 3$  is the sliding exponent, and  $C_b$  is a dimensional basal-friction coefficient with units MPa (m a<sup>-1</sup>)<sup>-1/3</sup>. A Weertman law is a simplified hard-bed parameterization and does not explicitly represent processes such as effective-pressure dependence, cavitation, or deforming till. We hold the basal law fixed so that the experiment tests the proposed mapping with one consistent ice-flow model.

We prescribe observed velocities along the non-terminus boundaries and apply an ocean-pressure traction condition at the termini. Positivity and the grounding transition are imposed through

$$C_b = \phi C_0 e^C, \quad (3)$$

where  $C$  is the dimensionless inversion control used to represent spatial variation in basal friction and  $C_0 = 0.01$  MPa (m a<sup>-1</sup>)<sup>-1/3</sup>.  $\phi$  is a height-above-flotation ramp that continuously reduces basal resistance as flotation is approached and enforces  $C_b = 0$  over floating ice (Leguy and others, 2014; Seroussi and others, 2014). Appendix section Ice-flow equations, grounding transition, and inversion method describes the full stress balance and grounding transition.



**Fig. 3.** Sector-wide inversion that provides the reference  $C$ : (a) observed speed, (b) the inversion-reference  $C$ , and (c) the magnitude of the difference between modeled and observed velocity.

The inversion minimizes the total objective function

$$J(\mathbf{u}, C; r_C) = \frac{1}{2NU^2} \sum_{i=1}^N \|\mathbf{u}(\mathbf{x}_i) - \mathbf{u}_{\text{obs}}(\mathbf{x}_i)\|^2 + \frac{1}{2|\Omega|} \left( \frac{L}{r_C} \right)^2 \int_{\Omega} \|\nabla C\|^2 dA, \quad (4)$$

where  $N$  is the number of paired-velocity observations,  $|\Omega|$  is the domain area,  $U = 1 \text{ m a}^{-1}$  and  $L = 7.5 \text{ km}$  are fixed reference scales that make both terms dimensionless, and  $r_C$  controls smoothness. Smaller  $r_C$  gives stronger smoothing. The misfit is unweighted, matching the velocity root-mean-square error (RMSE) that we use for evaluation. Following the use of L-curves in ice-sheet inverse models (Wolovick and others, 2023), we selected  $r_C \simeq 0.014$  using a maximum-curvature criterion (Appendix Fig. 8). We obtain gradients of Eqn 4 using the adjoint method. Only  $C$  is inverted; the rate factor in Glen’s flow law is prescribed from the englacial temperature field of Van Liefferinge and Pattyn (2013). We use the resulting inversion friction control field (Fig. 3b) for every learning experiment.

## Dataset and predictor configurations

We first interpolate every predictor product to the control points of the model mesh, where the inversion also defines  $C$ , and compute the gradients and derived predictors from these finite-element fields. We then evaluate the fields at the 450 m MEaSUREs pixel centers to construct the dataset, with each row containing the predictors and inversion-reference  $C$  at one pixel center. The predictors are finite-element fields on the mesh, so they cannot resolve variation finer than the mesh: each central 50 km square contains 169–330 control points, whereas the 500 m BedMachine grid has about 10,000 cells in the same area; the

**Table 1.** Sources and grid spacing of the predictors. Cells per square is the number of source-grid cells in a 50 km square,  $(50 \text{ km} / \text{grid spacing})^2$ , before any masking; neighboring cells are not independent, because the products are interpolated or smoothed. Gradients have the resolution limits of the fields they are derived from.

| Predictor(s)                  | Product/source                                     | Grid spacing           | Cells per square | Interpretation in this study                                                                                                                   |
|-------------------------------|----------------------------------------------------|------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| $s, h, b$ and their gradients | BedMachine<br>Antarctica v2                        | 500 m                  | $\sim 10,000$    | Geometry products. Fast-flow bed reconstruction uses mass conservation and velocity, and is not a direct measurement of local basal processes. |
| $\tau_d$                      | Derived from<br>BedMachine $h, s$                  | 500 m source<br>fields | $\sim 10,000$    | Driving-stress estimate derived from thickness and surface slope.                                                                              |
| $T_s$                         | Quantarctica surface-<br>air-temperature<br>raster | 5 km                   | $\sim 100$       | Large-scale surface thermal boundary, not a direct observation of basal temperature.                                                           |
| $Q_b$                         | Quantarctica<br>geothermal-heat-flux<br>raster     | 5 km                   | $\sim 100$       | Model-derived regional basal heat-flux estimate; interpolation does not add fine-scale information.                                            |
| $m_a$                         | ADMAP-2S<br>EPSG:3031 grid                         | 5 km                   | $\sim 100$       | Magnetic anomaly containing broad information about crustal magnetization; it does not uniquely identify basal material.                       |
| $g_d$                         | AntGG2021 oriented<br>EPSG:3031 grid               | 5 km                   | $\sim 100$       | Gravity disturbance containing information about subsurface density; it does not uniquely identify rock type or subglacial water.              |
| $a_{bs}$                      | Derived from<br>BedMachine $b, s$                  | 500 m source<br>fields | $\sim 10,000$    | Relative alignment of two geometry gradients; does not use flow direction.                                                                     |

other products are gridded at 5 km, about 100 cells per square, and are coarser than the mesh (Table 1). The trained MLPs are applied at the control points to produce the predicted  $C$  fields used in the forward simulations. Eligible rows must have finite values for all predictors and both velocity components, with  $\phi > 0.1$  to exclude floating ice and the immediate grounding transition. We use the same eligibility criteria for all predictor configurations and experiments. Appendix Figure 9 shows this eligible region mapped against the full modeled domain. Table 2 summarizes the twelve predictors and their six configurations (CFG01–CFG06). Appendix section Eligible rows, predictor construction, and support provides gradient construction and exact row counts.

The one additional predictor in CFG06 relative to CFG05 measures bed–surface gradient alignment,

$$a_{bs} = \frac{\nabla b \cdot \nabla s}{\sqrt{(\|\nabla b\|^2 + \epsilon_b^2)(\|\nabla s\|^2 + \epsilon_s^2)}}, \quad (5)$$

where  $\epsilon_b = 5.4 \times 10^{-3}$  and  $\epsilon_s = 1.5 \times 10^{-4}$  are dimensionless slope scales. When both gradients are large relative to these values,  $a_{bs}$  approaches  $\cos \theta_{bs}$ . The added terms prevent the alignment from becoming unstable where either gradient is very small. This predictor describes relative alignment without using velocity direction. The surface gradient defines the local driving-stress direction, so comparing it with the bed gradient measures whether bed slopes are oriented along or across the driving direction, without using observed velocity. The choice of  $\epsilon_b$  and  $\epsilon_s$  is described in Appendix section Eligible rows, predictor construction, and support.

CFG01 contains all ice-product predictors and CFG02 the selected ice-product subset. CFG03 contains all subglacial-product predictors and CFG04 the selected subglacial-product subset. CFG05 combines the two selected subsets, and CFG06 adds Eqn 5. We selected the subsets used in CFG02 and CFG04 from preliminary catchment-scale analyses of modeled velocity conducted before the present spatially withheld experiments, and we did not modify them based on the square results.

## Spatially withheld square experiments and predictor support

The primary experiment uses ten 50 km squares distributed across the study region (Fig. 2b). For each square, we exclude the full 130 km footprint from the training and validation datasets, leaving a 40 km buffer on each side of the central evaluation square. The ten withheld regions do not overlap. Following the rationale of spatially separated evaluation (Roberts and others, 2017; Valavi and others, 2019), the

**Table 2.** Predictors in the six configurations. Ice-product predictors are denoted I and subglacial-product predictors S. A filled circle indicates inclusion.

| Symbol         | Predictor                      | Unit               | Group | CFG01 | CFG02 | CFG03 | CFG04 | CFG05 | CFG06 |
|----------------|--------------------------------|--------------------|-------|-------|-------|-------|-------|-------|-------|
| $s$            | surface elevation              | m                  | I     | •     | •     |       |       | •     | •     |
| $h$            | ice thickness                  | m                  | I     | •     | •     |       |       | •     | •     |
| $\ \nabla s\ $ | surface-gradient magnitude     | 1                  | I     | •     | •     |       |       | •     | •     |
| $\ \nabla h\ $ | thickness-gradient magnitude   | 1                  | I     | •     | •     |       |       | •     | •     |
| $\tau_d$       | driving stress                 | MPa                | I     | •     |       |       |       |       |       |
| $T_s$          | surface air temperature        | °C                 | I     | •     | •     |       |       | •     | •     |
| $b$            | bed elevation                  | m                  | S     |       |       | •     | •     | •     | •     |
| $\ \nabla b\ $ | bed-gradient magnitude         | 1                  | S     |       |       | •     | •     | •     | •     |
| $Q_b$          | geothermal heat flux           | mW m <sup>-2</sup> | S     |       |       | •     | •     | •     | •     |
| $g_d$          | gravity disturbance            | mGal               | S     |       |       | •     |       |       |       |
| $m_a$          | magnetic anomaly               | nT                 | S     |       |       | •     |       |       |       |
| $a_{bs}$       | bed-surface gradient alignment | 1                  | S     |       |       |       |       |       | •     |

buffer reduces dependence on nearby spatially correlated training rows. It does not remove the dependence introduced by the common sector-wide inversion or by non-local ice flow. The central 50 km square is the primary evaluation area; the surrounding buffer and complete footprint are used only for maps and secondary summaries, not as additional independent tests. We chose the 40 km buffer before model fitting from the spatial correlation of observed ice speed, which falls to about 0.1 at 40 km separation (Appendix section Spatial separation and buffer choice).

We selected candidate square locations using spatial separation and predictor-support criteria so that the remaining training region generally represented the withheld predictor values well. We assessed support using both marginal and joint criteria. Marginal support requires every included predictor at a held-out point to lie within the 1st–99th percentile range of the corresponding eligible non-held-out rows. Joint support asks whether the complete predictor combination has a sufficiently close analog in the remaining training region. We report the area fraction satisfying both criteria; support is not used to select which points are evaluated. Details of the joint-support transformation and distance threshold are given in Appendix section Eligible rows, predictor construction, and support. A separate regional experiment withholds the two inter-catchment corridors and is reported in Appendix section Basal-friction control: distributions and comparisons.

We used no criterion based on the inversion-reference  $C$  or modeled velocity errors to select squares. We examined the distribution of inversion-reference  $C$  only afterward because the reference field is a regularized, non-unique inversion solution, and a different predicted  $C$  may still reproduce the observed velocity. We did not use the inversion-reference  $C$  or modeled velocity error to select the squares, so their locations do not depend on how well the predicted  $C$  performs there.

### *Catchment-scale transfer test*

In addition to the ten spatial tests, we use PIG as a catchment-scale transfer test chosen before we examined its results. This tests whether an MLP can transfer a point-wise relationship learned elsewhere in the Amundsen Sea sector across an entire glacier catchment. We excluded all eligible PIG rows from the training and validation dataset; the remaining grounded sector, including Thwaites, Dotson–Crosson, and the two inter-catchment corridors, provided the training and validation rows. We evaluated predictor support in PIG using the same marginal and joint criteria as the square experiments, without using  $C$  or velocity errors.

## **Model fitting and repeated realizations**

We use an MLP as a flexible nonlinear representation of  $f(\cdot)$  in Eqn 1 to test whether inversion-derived friction can be expressed as a local function of the available predictors. An MLP is a neural network in which each layer forms weighted sums of its inputs and passes them through a nonlinear function; stacking several layers lets it represent nonlinear relationships without a prescribed form. The model uses only the predictor values at each location. Approaches that also use neighboring values, such as convolutional or graph neural networks, would test a different hypothesis: that  $C$  depends on the spatial pattern of the predictors around a location, and not only on their local values. The MLP contains ten hidden layers of 200 units and predicts the reference  $C$  using the predictors in each configuration. We robustly scale inputs and targets using medians and interquartile ranges calculated from the training data only. Appendix section MLP training provides further details on the network implementation, training procedure, repeated realizations, and training and validation histories.

For each square experiment, we reproducibly split the eligible rows remaining after removing the excluded footprint into 90% for training and 10% for validation. Nearby training and validation rows are not statistically independent because of interpolation, predictor resolution, spatial autocorrelation, and

regularization of the inversion-reference field. We therefore use the random split only for model fitting and selection, and evaluate performance separately in the spatially withheld central square. Appendix section Eligible rows, predictor construction, and support provides exact row counts.

For each experiment and predictor configuration, we train ten reproducible realizations that differ in the training/validation split, initial weights, and epoch shuffling. Using multiple realizations reduces sensitivity to any one set of training choices and provides a more stable estimate of the predicted  $C$  field (Hansen and Salamon, 1990; Breiman, 1996). Each realization uses the same training and validation rows in every predictor configuration, so the configurations can be compared directly. The differences among the ten realizations show how much the predicted  $C$  depends on these training choices; they are not an estimate of prediction uncertainty. The prediction for each experiment and configuration is the median of the ten predicted  $C$  fields at each mesh control point.

## Forward simulations and measures of performance

We evaluate each median predicted  $C$  field in a separate Icepack forward simulation. For each square, predicted  $C$  replaces the inversion-reference  $C$  only in the grounded, eligible part of the 130 km withheld region. The inversion-reference  $C$  is retained everywhere outside that footprint. The PIG experiment uses the same procedure over the withheld catchment. Appendix section Sensitivity to the replacement region compares this design with replacing  $C$  across the whole sector. At the edge of the replaced region, the control changes steeply but continuously to the inversion-reference  $C$  within the mesh cells that cross the edge; we applied no additional smoothing. All forward simulations converged within 50 Newton iterations.

The primary performance measure is vector velocity RMSE (hereafter velocity RMSE) against the original paired MEaSUREs velocity components at their regular 450 m pixel centers over the evaluation area  $A$ . We evaluate modeled velocity at the same locations. The inversion misfit (Eqn 4) is computed at these same pixel centers. This evaluation is distinct from the mesh-interpolated observation field that the solver uses for boundary conditions and as its initial velocity. The spatial average below is approximated using the mean squared vector residual at these equally spaced points:

$$\text{RMSE}_u = \left[ \frac{\int_A \|\mathbf{u}_{\text{ML}} - \mathbf{u}_{\text{obs}}\|^2 dA}{\int_A dA} \right]^{1/2}. \quad (6)$$

The subscript ML denotes the simulation using the MLP-predicted  $C$ .

For the maps, we also use the local vector-error magnitude

$$e(\mathbf{x}) = \|\mathbf{u}(\mathbf{x}) - \mathbf{u}_{\text{obs}}(\mathbf{x})\|.$$

263 The spatially uniform- $C$  baseline uses a single control value with no spatial variation from the predictors.  
 264 This value is the finite-element-area-weighted mean inversion-reference  $C$  over the grounded cells ( $\phi > 0.1$ )  
 265 of the training and validation region and is applied uniformly within the withheld region. Because  $C_b \propto$   
 266  $\exp(C)$ , averaging  $C$  corresponds to a geometric mean of  $C_0 \exp(C)$  before multiplication by the grounding  
 267 ramp  $\phi$ . We apply it over the same withheld replacement region as the MLP-predicted field, retaining  
 268 the inversion-reference  $C$  outside that region in both simulations, so the two simulations differ only in  
 269 the spatial pattern of  $C$  inside that region. The comparison therefore asks whether spatial variation in  $C$   
 270 inferred from the predictors improves modeled velocity beyond what a single representative friction control  
 271 provides. We also report  $\text{RMSE}_{\text{relative}} = \text{RMSE}_{\text{ML}}/\text{RMSE}_{\text{uniform}}$ ; values below one indicate improvement  
 272 over the uniform- $C$  baseline.

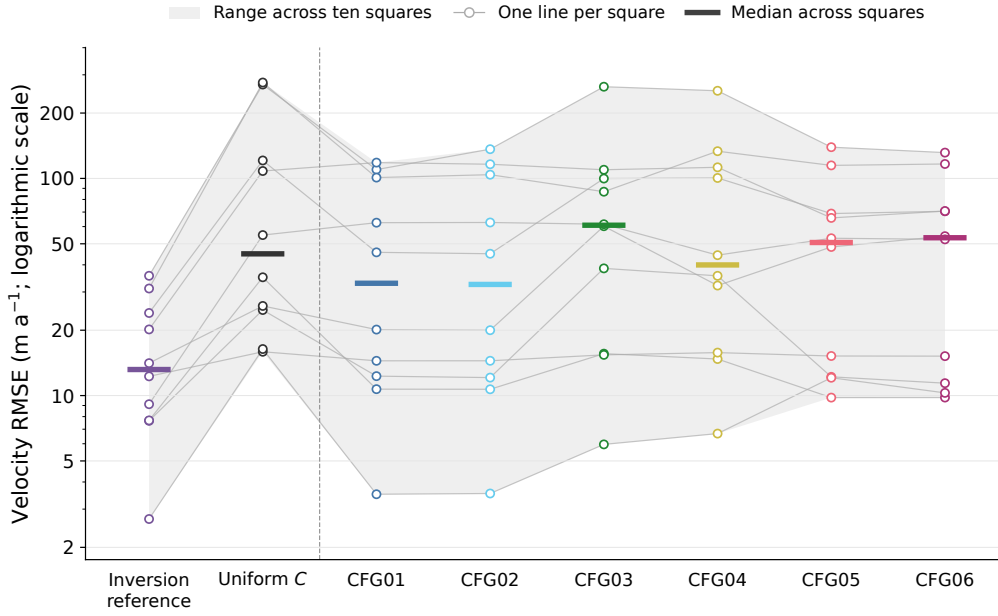
273 The ten held-out squares are the independent spatial tests. For each predictor configuration, we summa-  
 274 rize performance by the median and interquartile range of the ten square RMSE values. We also compare  
 275 configurations that differ only by added predictors square by square, counting how many squares each  
 276 performs better in.

## 277 RESULTS AND DISCUSSION

### 278 Modeled velocity performance

279 Figure 4 shows the velocity RMSE in each of the ten central squares for the six predictor configurations,  
 280 together with the uniform- $C$  baseline and the reference- $C$  best case. The predicted  $C$  often improves  
 281 modeled velocity relative to the uniform- $C$  baseline, but performance varies substantially among squares.

282 In Table 3, the median and interquartile range describe the typical velocity RMSE of each control  
 283 across the ten squares. The improvement count is the number of squares in which a configuration has  
 284 lower RMSE than uniform  $C$  in that same square, and the relative RMSE is the configuration's RMSE  
 285 divided by the uniform- $C$  RMSE in the same square. The two ice-product configurations have the lowest  
 286 median RMSE, 32.9 m a<sup>-1</sup> for CFG01 and 32.5 m a<sup>-1</sup> for CFG02, and each improves on uniform  $C$  in eight  
 287 of ten squares. CFG03 and CFG04, the all and selected-subglacial configurations, improve on uniform  $C$



**Fig. 4.** Velocity RMSE against observed MEaSUREs velocity in the ten central squares. The inversion reference and square-specific uniform- $C$  baseline are followed by forward simulations using the median predicted  $C$  for all six configurations. Each gray line connects the same square; shading spans the minimum to maximum across squares, and short horizontal marks show the medians. The vertical axis is logarithmic. We calculate uniform  $C$  separately from the eligible region outside each withheld region; both the predicted and uniform controls replace the inversion reference only inside that region. The inversion reference was fitted using sector-wide observations, including those in the test squares.

in six and seven squares, respectively, while CFG05 and CFG06, which combine the selected ice-product and subglacial-product predictors, improve on uniform  $C$  in nine squares.

The two kinds of summary can differ. The median is the average of the fifth- and sixth-ranked squares, and each control is ranked separately. For uniform  $C$ , the middle squares are SQ04 ( $35.1 \text{ m a}^{-1}$ ) and SQ09 ( $54.8 \text{ m a}^{-1}$ ). CFG05 lowers RMSE in nine squares and in SQ04 reduces it to  $9.8 \text{ m a}^{-1}$ , which moves SQ04 to the bottom of its ranking. Its middle squares are then SQ07, the only square where CFG05 is worse than uniform  $C$  ( $48.3$  against  $25.9 \text{ m a}^{-1}$ ), and SQ09 ( $53.0 \text{ m a}^{-1}$ ). The median RMSE of CFG05 is therefore higher than that of uniform  $C$  even though CFG05 improves on it in nine of ten squares. The same holds for CFG06 and CFG03, which improve on uniform  $C$  in nine and six squares, respectively. The inversion-reference solution has a substantially lower median RMSE of  $13.2 \text{ m a}^{-1}$ . This is not a withheld benchmark, because the inversion was fitted using velocity observations from the complete sector, including the evaluation squares. It nevertheless provides context for the velocity fit obtained with the spatially resolved inversion. The substantially larger errors from the predicted  $C$  fields show that the point-wise mappings recover only part of the improvement available from the inversion-derived spatial structure. At the same time, their frequent improvement over uniform  $C$  shows that the predictors contain

**Table 3.** Velocity RMSE against observed MEaSURES velocity in the ten central squares. Values are the median and interquartile range (IQR) across squares; the median RMSE in each row is taken over that row’s own ten squares and is not a paired comparison. Relative RMSE and the improvement counts compare each configuration with its square-specific uniform- $C$  baseline within the same square. RMSE is in  $\text{m a}^{-1}$ .

| Control/config.     | Predictors           | Median RMSE | IQR        | Better than uniform | Median relative RMSE | Relative RMSE IQR |
|---------------------|----------------------|-------------|------------|---------------------|----------------------|-------------------|
| Uniform $C$         | spatially uniform    | 44.9        | 25.0–117.8 | –                   | 1.00                 | 1.00–1.00         |
| Inversion reference | –                    | 13.2        | 8.0–23.0   | 10/10               | 0.19                 | 0.17–0.29         |
| CFG01               | all ice              | 32.9        | 12.8–91.3  | 8/10                | 0.45                 | 0.37–0.88         |
| CFG02               | selected ice         | 32.5        | 12.7–93.7  | 8/10                | 0.50                 | 0.37–0.87         |
| CFG03               | all subglacial       | 60.9        | 21.3–96.6  | 6/10                | 0.97                 | 0.54–1.09         |
| CFG04               | selected subglacial  | 40.0        | 19.9–109.5 | 7/10                | 0.88                 | 0.56–1.03         |
| CFG05               | selected combined    | 50.7        | 13.0–68.1  | 9/10                | 0.59                 | 0.50–0.90         |
| CFG06               | combined + alignment | 53.3        | 12.4–70.7  | 9/10                | 0.61                 | 0.47–0.88         |

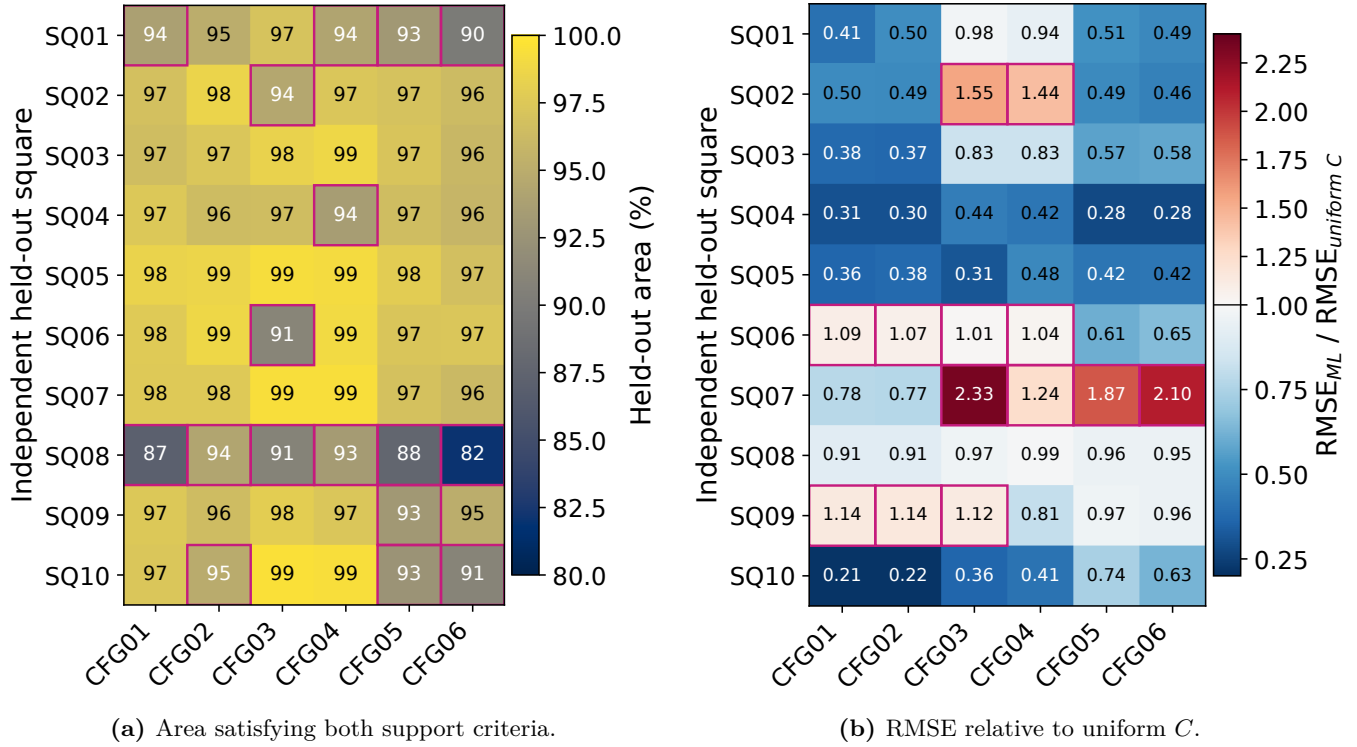
information relevant to the effective friction control.

Removing driving stress from CFG01 to form CFG02 changes the median velocity RMSE only slightly, from 32.9 to 32.5  $\text{m a}^{-1}$ , and both configurations improve on uniform  $C$  in eight squares. Driving stress is derived from ice thickness and surface slope, which CFG02 already includes, so supplying it explicitly provides no consistent improvement across the ten squares. Among the subglacial configurations, CFG04 has a lower median velocity RMSE than CFG03 (40.0 versus 60.9  $\text{m a}^{-1}$ ), but CFG04 has lower velocity RMSE than CFG03 in only five of the ten squares; CFG03 performs better in the other five. Relative to uniform  $C$ , CFG03 reduces velocity RMSE in six squares and CFG04 in seven.

Combining the selected predictor groups does not consistently outperform either group alone. CFG05 performs better than CFG02 in three squares and better than CFG04 in seven. Adding bed–surface gradient alignment in CFG06 again divides the comparison evenly, with CFG06 performing better than CFG05 in five squares and worse in five. No configuration is better than another in every square (Fig. 5b), so the ten squares do not identify a single best configuration; this does not negate the improvements over uniform  $C$ . Overall, the relative performance of the predictor configurations changes among squares, indicating that no tested predictor set provides a uniformly successful point-wise relationship across the study region.

### Predictor support and modeled velocity performance

We next examine whether poor modeled velocity performance is associated with low predictor support. Across the ten primary squares, predictor support is generally high (Fig. 5a). Marginal support covers more



**Fig. 5.** Predictor support and modeled velocity performance in the ten spatial tests for six predictor configurations. Panel (a) gives the percentage of each held-out area satisfying both marginal and joint support (values rounded); magenta outlines mark values below 95%. Panel (b) gives relative RMSE; magenta outlines mark cases that do not improve on the spatially uniform- $C$  baseline. Marginal bounds use eligible non-held-out rows, while joint matches use the 5 km reference grid within the training region.

than 94% of each primary square, while lower joint support is concentrated mainly in SQ08. However, high support does not guarantee good modeled velocity performance. Figure 5b shows performance relative to uniform  $C$ ; a relative RMSE greater than one means that the predicted control performs worse than the uniform- $C$  baseline. SQ07-CFG03, for example, has 99.4% support by both criteria but a relative RMSE of 2.33, while SQ02-CFG04 has 97.3% support and a relative RMSE of 1.44 (Table 4). These failures cannot be explained solely by individual predictors falling outside their training ranges or by the absence of a nearby multivariable match.

Predictor support indicates whether similar predictor combinations are represented in the training region, but does not distinguish between densely and sparsely sampled parts of predictor space. We therefore examine whether predictor combinations with fewer similar training rows, that is, lower training density, are associated with errors in the predicted  $C$  for CFG02 and CFG04 (Appendix section Training density and prediction error). We measure training density by the distance from each held-out row to its 20th-nearest training row in predictor space, expressed as a percentile  $p$  of the same distance among

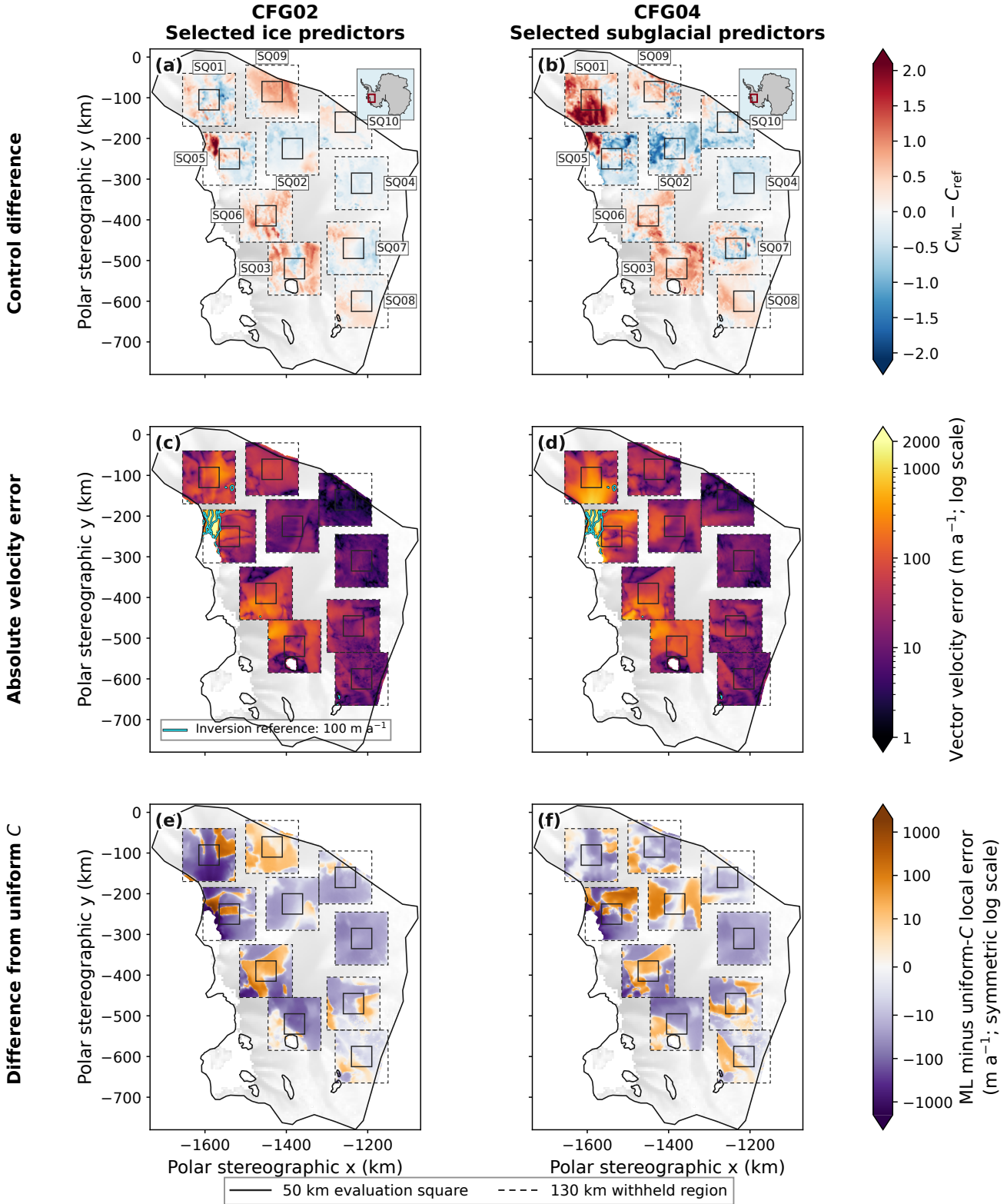
**Table 4.** Examples of poor performance, defined here as velocity RMSE exceeding the uniform- $C$  baseline, despite high predictor support. Support is the percentage of held-out points that satisfy both the marginal and joint criteria.

| Square               | Config. | Support (%) | ML RMSE | Uniform RMSE | Relative RMSE |
|----------------------|---------|-------------|---------|--------------|---------------|
| (m a <sup>-1</sup> ) |         |             |         |              |               |
| SQ02                 | CFG04   | 97.3        | 35.6    | 24.8         | 1.44          |
| SQ06                 | CFG01   | 97.6        | 118.3   | 108.2        | 1.09          |
| SQ07                 | CFG03   | 99.4        | 60.2    | 25.9         | 2.33          |
| SQ09                 | CFG02   | 96.4        | 62.7    | 54.8         | 1.14          |

training rows, so that a high  $p$  means few similar training rows. For CFG02,  $C$  RMSE increases as training density decreases in seven of the ten squares, but the overall relationship is weak. The median within-square Spearman correlation between  $p$  and absolute  $C$  error is only 0.069, and the median  $C$  RMSE differs by only 0.004 across the three training-density categories. CFG04 shows no similar pattern, with a median correlation between  $p$  and absolute  $C$  error of  $-0.052$ . Low training density is therefore associated with larger  $C$  RMSE in some CFG02 experiments, but the relationship is weak and is not shared by CFG04. Training density also does not consistently correspond to modeled velocity performance, as illustrated by the square-level examples in Appendix section Training density and prediction error. In some squares, modeled velocity has lower RMSE than uniform  $C$  in low-density categories but not in high-density categories. Training density therefore does not by itself explain the failures to improve modeled velocity relative to uniform  $C$ .

### Spatial patterns of success and failure

Figure 6 compares CFG02, the selected ice-product configuration, with CFG04, the selected subglacial-product configuration. The two ice-product configurations have nearly identical summary performance, so CFG02 represents that predictor group without repeating a visually similar map, and CFG04 represents the subglacial predictor group. For each square, we trained a separate set of MLPs, excluding the full 130 km footprint from training and validation; the panels therefore assemble ten separately withheld predictions rather than one sector-wide prediction. In every simulation, the predicted and uniform controls replace the inversion reference only inside the corresponding footprint. Across the central 50 km squares, CFG02 has lower velocity RMSE than CFG04 in eight of ten tests. Over the complete 130 km footprints, both configurations improve on uniform  $C$  in nine of ten tests, with median relative RMSEs of 0.64 for CFG02 and 0.86 for CFG04. Across the ten central 50 km squares, velocity error exceeds  $100 \text{ m a}^{-1}$  over 13.0%



**Fig. 6.** Spatial comparison of the selected ice-product configuration (CFG02; left) and selected subglacial-product configuration (CFG04; right). Panels (a, b) show  $C_{ML} - C_{ref}$ . Panels (c, d) show the vector velocity error relative to MEaSUREs observations,  $e_{ML} = \|\mathbf{u}_{ML} - \mathbf{u}_{obs}\|$ . Panels (e, f) show  $e_{ML} - e_{uniform}$ ; negative values indicate lower local error for the predicted control and positive values indicate lower error for uniform  $C$ . Cyan contours in panels (c, d) mark an inversion-reference velocity error of  $100 m a^{-1}$ . In each experiment, both controls replace the inversion reference only inside the complete 130 km footprint. Velocity errors use the original paired MEaSUREs components and are shown as  $1.8 km$  bin means for legibility; we calculated all statistics from the evaluated observation-grid points.

for CFG02 and 22.8% for CFG04, compared with only 0.13% for the inversion-reference solution. Thus, although the predicted controls often improve on uniform  $C$ , regions of large absolute velocity error remain much more extensive than for the fitted inversion-reference control, as also observed in Table 3.

The maps show that the ice-product configuration provides an advantage across several distinct flow settings. In the fast PIG test SQ01, the central-square RMSE ratio is 0.50 for CFG02 and 0.94 for CFG04, with the largest CFG02 improvement concentrated in the fast-flowing part of the square. The local error difference adds an important distinction: CFG02 reduces error over 84.8% of the central square, with a median pointwise reduction of  $133.5 \text{ m a}^{-1}$ , whereas CFG04 reduces error at every evaluated point but has a much smaller median pointwise reduction of  $14.8 \text{ m a}^{-1}$ . Thus, the fraction of the square with lower local error does not determine the RMSE improvement; the magnitude and spatial distribution of the error reductions also matter. CFG02 also gives the larger improvement in moderately fast Thwaites flow in SQ03 (0.37 versus 0.83). There, local error is lower than for uniform  $C$  over 90.4% of the central square for CFG02 and 77.5% for CFG04. Good performance is not confined to fast flow: both configurations improve substantially on uniform  $C$  in the slow Thwaites square SQ04 (0.30 and 0.42) and the slow PIG–Thwaites corridor square SQ10 (0.22 and 0.41).

The large absolute residuals in the SQ05 footprint occur where it extends from the PIG–Thwaites corridor into fast PIG flow. These large residuals contribute disproportionately to the footprint-wide RMSE, yet both learned controls still improve on uniform  $C$  (ratios of 0.85 and 0.86). Figure 6 panels (e, f) show that this footprint contains both reductions and increases in local error: CFG02 reduces error over 80.7% of its evaluated area and CFG04 over 60.2%. The large absolute errors in panels (c, d) therefore coexist with improvements relative to the uniform control. The inversion-reference velocity error is also elevated in this part of SQ05, indicating that some of the mismatch remains even with the fitted reference  $C$ . Over the complete footprint, however, the velocity RMSEs are  $539.5$  and  $546.2 \text{ m a}^{-1}$  for CFG02 and CFG04 respectively, compared with  $77.1 \text{ m a}^{-1}$  for the inversion reference. The predicted controls therefore introduce substantial additional error over the reference control.

Although CFG02 performs better than CFG04 in most squares, the relative performance of the two predictor groups varies by location. In SQ02, which spans the PIG–Thwaites corridor and Thwaites, CFG02 performs substantially better than the uniform- $C$  baseline, whereas CFG04 performs worse. The central-square RMSE ratios are 0.49 for CFG02 and 1.44 for CFG04, and the corresponding full-footprint ratios are 0.58 and 1.91. The local comparison shows that CFG02 reduces error over 99.8% of the central

square, whereas CFG04 reduces it over only 42.5%. The higher CFG04 RMSE therefore accompanies a mixture of local improvements and deteriorations, rather than uniformly larger errors. CFG02 also agrees more closely with the inversion-reference control in the central square, with a  $C$  RMSE of 0.22 compared with 0.61 for CFG04. SQ09, spanning PIG and the PIG–Thwaites corridor, shows the opposite pattern: CFG04 improves on uniform  $C$  while CFG02 does not, with central-square ratios of 0.81 and 1.14 and full-footprint ratios of 0.88 and 1.20, respectively. In this central square,  $e_{\text{ML}} - e_{\text{uniform}}$  is positive at every evaluated point for CFG02 and negative at every evaluated point for CFG04. CFG04 also has the lower central-square  $C$  RMSE in SQ09 (0.49 versus 0.74). These examples show that neither predictor group performs better at every location.

In SQ06, both configurations are slightly worse than uniform  $C$  in the central Thwaites square, with relative RMSEs of 1.07 for CFG02 and 1.04 for CFG04, despite simultaneous predictor support exceeding 98%. Local error is lower than uniform  $C$  over only 12.6% of the central square for CFG02 and 22.6% for CFG04. Over the complete footprint the RMSE ratios decrease to 0.62 and 0.79, and local error reductions cover 54.1% and 59.5% of the evaluated area. CFG04 changes in the same direction in SQ07, from 1.24 in the central square to 0.80 over the complete footprint. In other squares the result does not depend on the area evaluated: the CFG04 failure in SQ02 and the CFG02 failure in SQ09 remain over the complete footprints, with ratios of 1.91 and 1.20. The complete footprint includes the 40 km buffer, which lies closer to the training rows than the central square does, but the change between the two areas does not follow one direction: across the ten squares and these two configurations, the complete-footprint relative RMSE is lower than the central-square value in nine cases and higher in eleven. The complete footprint also covers a larger and more varied area than the central square, so the comparison depends on the ice included rather than on distance from the training rows.

SQ08 presents a different case: the uniform- $C$  error is already only  $15.9 \text{ m a}^{-1}$ , compared with  $12.3 \text{ m a}^{-1}$  for the inversion-reference simulation, leaving relatively little scope for improvement. CFG02 and CFG04 give RMSE values of 14.5 and  $15.8 \text{ m a}^{-1}$ , respectively. Although local error is lower than for uniform  $C$  over 75.0% and 71.3% of the central square, the median local reductions are only 4.3 and  $5.0 \text{ m a}^{-1}$ . In this case, widespread negative values in panels (e, f) represent small absolute gains, and the larger residuals elsewhere limit the RMSE reduction.

As discussed above, agreement with the inversion-reference  $C$  and velocity performance coincide in SQ02 and SQ09. In fact, when comparing CFG02 with CFG04 within each of the ten central squares, the

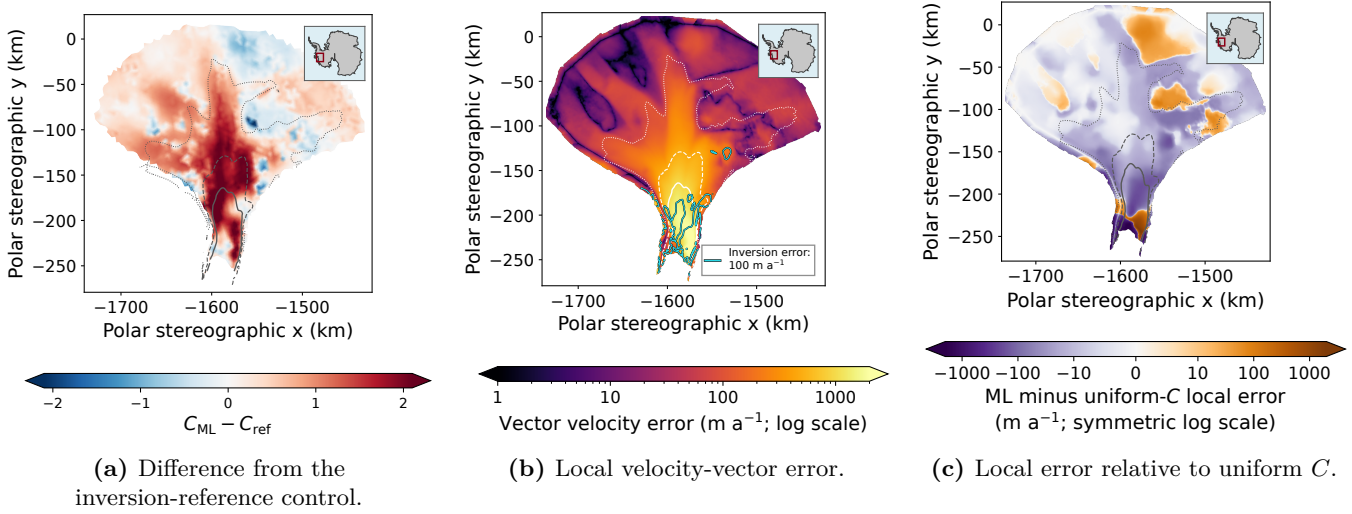
configuration with lower  $C$  RMSE also has lower velocity RMSE. However, the size of the improvement in velocity RMSE does not follow directly from the size of the difference in  $C$  RMSE, because the  $C$  RMSE weights all locations equally whereas the velocity RMSE is dominated by faster flow. Over the complete SQ06 footprint, for example, CFG02 and CFG04 have nearly identical  $C$  RMSE values (0.435 and 0.440), but their velocity RMSE ratios differ substantially (0.62 and 0.79). CFG02 has the smaller  $C$  error in flow of 100–1000 m a<sup>-1</sup>, which carries most of the velocity error (91% of the squared error for CFG04), and CFG04 has the smaller  $C$  error in slower ice, which carries only 4%. Closer overall agreement with the inversion-reference  $C$  therefore does not translate into a proportional improvement in velocity RMSE. Within the individual squares, larger absolute local differences from the inversion-reference  $C$  generally coincide with larger local velocity errors, although the strength of this relationship varies by location. The median within-square Spearman correlations are 0.32 for CFG02 and 0.42 for CFG04, indicating a modest positive association rather than a one-to-one correspondence between  $C$  error and velocity error, consistent with a given  $C$  error producing a much smaller velocity error in slow ice than in fast flow.

Taken together, the ten central-square tests show that the selected ice-product and subglacial-product configurations improve velocity RMSE relative to uniform  $C$  in most squares. Neither configuration improves every square, and some failures persist across the complete withheld footprint despite high predictor support. The selected predictors therefore contain information that can improve modeled velocity relative to a uniform control, but they do not define a point-wise relationship that is consistently successful across locations.

### Catchment-scale test at PIG

The secondary catchment-scale experiment tests whether the point-wise relationship remains useful when all of PIG is withheld. We evaluate CFG02, the selected ice-product configuration, using the same ice-flow model and evaluation procedure as in the square experiments. The spatially uniform- $C$  baseline produces a vector velocity RMSE of 384.9 m a<sup>-1</sup>, whereas CFG02 produces an RMSE of 354.8 m a<sup>-1</sup>. This corresponds to a modest 8% reduction in RMSE relative to the uniform- $C$  baseline. The inversion-reference simulation has an RMSE of 48.1 m a<sup>-1</sup>, showing that the transferred point-wise mapping recovers only a small fraction of the improvement obtained with the spatially resolved inversion.

CFG02 lowers the local velocity error relative to uniform  $C$  over 77.6% of the evaluated PIG area (Fig. 7). The improvement occurs across all observed-speed classes. The relative RMSE is 0.59 below



**Fig. 7.** Spatial evaluation of CFG02 with all of PIG withheld. Panel (a) shows the median predicted control relative to the inversion-reference control. Panel (b) shows  $\|\mathbf{u}_{\text{ML}} - \mathbf{u}_{\text{obs}}\|$ . Panel (c) shows the difference between the local CFG02 and uniform- $C$  velocity errors; negative values indicate lower error for CFG02 and positive values indicate lower error for uniform  $C$ . Contours mark observed speeds of 100 (dotted), 500 (dashed), and 1000  $\text{m a}^{-1}$  (solid). The cyan contour in panel (b) marks an inversion-reference velocity error of 100  $\text{m a}^{-1}$ . Velocity fields are rendered from the original 450 m grid and smoothed for display; all reported statistics use the original grid values. Both CFG02 and uniform  $C$  replace the inversion reference only inside the PIG holdout.

100  $\text{m a}^{-1}$ , 0.84 between 100 and 500  $\text{m a}^{-1}$ , 0.89 between 500 and 1000  $\text{m a}^{-1}$ , and 0.96 above 1000  $\text{m a}^{-1}$ . The improvement is therefore spatially widespread, but its influence on the catchment-wide statistic is concentrated in fast flow. The fastest 10% of the evaluated area, corresponding to observed speeds above approximately 504  $\text{m a}^{-1}$ , contributes 59.8% of the total squared-error reduction relative to uniform  $C$ , but still accounts for 89.5% of the CFG02 squared error. In the fastest flow ( $> 1000 \text{ m a}^{-1}$ ), the RMSE remains 1390.0  $\text{m a}^{-1}$ , compared with 1448.8  $\text{m a}^{-1}$  for uniform  $C$ . Thus, CFG02 improves much of the slow and intermediate-speed catchment interior and parts of the fast-flowing trunk, but provides only a modest relative improvement in the fastest downstream flow. This spatial concentration explains how CFG02 can lower error across most of the catchment while retaining a large catchment-wide RMSE.

Locations where the inversion-reference error is at least 100  $\text{m a}^{-1}$  comprise 3.9% of the evaluated PIG area but contain 47.1% of the CFG02 squared error. The remaining error is not confined to those locations: most locations with CFG02 error above 100  $\text{m a}^{-1}$  lie outside regions where the inversion-reference error exceeds 100  $\text{m a}^{-1}$ . Predictor support alone does not explain the spatial pattern. For CFG02, marginal support covers 97.5% of the evaluated PIG area, joint support covers 93.3%, and 91.9% satisfies both criteria. Of the net squared-error reduction relative to uniform  $C$ , 79.8% occurs within the area satisfying both support criteria. CFG02 nevertheless contains both successful and unsuccessful regions where predictor

support is strong. CFG02 has a  $C$  RMSE of 0.85 across PIG. Locations with larger absolute differences from reference  $C$  also tend to have larger velocity errors, with a Spearman correlation of 0.67. This indicates a relatively strong positive association and is larger than the median within-square correlation of 0.32 for CFG02. The association between the training-density percentile  $p$  and absolute  $C$  error is much weaker.  $C$  RMSE increases from 0.742 to 0.877 and 0.898 across the three categories of progressively lower training density, but the Spearman correlation between  $p$  and absolute  $C$  error is only 0.031. This near-zero correlation indicates that fewer nearby training examples do not consistently correspond to higher local  $C$  error across PIG.

### Implications for the point-wise hypothesis

Taken together, the experiments address three related questions: whether the predictor values in a held-out region are represented in the training data, whether the predicted  $C$  resembles the reference  $C$ , and whether the predicted  $C$  improves modeled velocity. The results show that these questions can have different answers. High predictor support can coexist with poor velocity performance, and lower training density does not consistently produce larger  $C$  errors. Agreement with the reference  $C$  is also related to, but does not determine, the resulting velocity error: CFG05, for example, has a median  $R_C^2$  of  $-1.41$  in the central squares yet improves on uniform  $C$  in nine of ten. Nor can the predictors, taken together, indicate in advance where the predicted  $C$  will improve on uniform  $C$ : a classifier trained on all twelve predictors is rarely better than predicting whichever outcome is more common in the withheld region (Appendix Predicting where transfer succeeds). The PIG experiment shows that the same distinctions remain important when the held-out region is expanded from a 50 km square to an entire glacier catchment.

The contrast between random and spatial holdouts bears directly on the hypothesis in Eqn 1. If  $C$  were a point-wise function of the predictors, a relationship that reproduces randomly withheld rows should reproduce spatially withheld rows with comparable accuracy wherever similar predictor values are well represented in the training data. CFG02 reproduces randomly withheld validation rows with a median  $C$  RMSE of 0.08 and  $R_C^2$  of 0.99 (Appendix Table 7), but in the withheld central squares its median  $C$  RMSE is 0.30 and  $R_C^2$  is negative in eight of ten (Appendix Table 8), although 94–99% of every square satisfies both support criteria (Fig. 5a). A fitted model can transfer poorly for other reasons, such as limited sampling of the predictor space or approximation error, but the errors do not decrease where the training data are denser: held-out rows whose training density is at least as high as that of a typical training row

have the same median  $C$  RMSE of 0.30 (Appendix Table 5). The difference is therefore not explained by the support and training-density measures tested here, and it points to structure in the inversion-derived  $C$  that the local predictors do not capture. Selecting the regularization or the stopping point against spatially withheld data might improve transfer, but a gain obtained by shrinking the prediction toward a spatially uniform control would reflect less use of the predictors rather than a closer point-wise relationship with them.

The spatial differences in performance are not simply explained by the BedMachine diagnostics either. Reported bed-elevation uncertainty has only a weak relationship with local velocity error across the square tests (Appendix BedMachine reconstruction and velocity error). Comparisons among BedMachine source classes are more limited because eight of the ten central squares contain only streamline-diffusion cells. Nevertheless, CFG02 velocity RMSE ranges from 3.5 to 136.3 m a<sup>-1</sup> across those eight squares, showing that differences in reconstruction class alone cannot explain the large spatial variation in performance. Reconstruction may still matter within a source class; for example, smoothing may remove local bed variations that could help distinguish basal settings, but the present analysis does not isolate this effect.

More fundamentally, the available predictors may not capture all of the physical variability relevant to  $C$ . Small-scale basal roughness can influence basal resistance (Hoffman and others, 2022). The observed velocity reflects that influence, so the inverted  $C$  contains it, but none of the predictors resolves roughness at these scales. Spatial variation in  $C$  caused by unresolved roughness therefore cannot be predicted from the available observables. The heat-flux, gravity, and magnetic products also provide indirect information at relatively coarse spatial resolution. Errors in the predictors can also hide a relationship that does exist. Where a predictor is poorly known, locations with the same bed conditions can have different predictor values, and locations with different conditions can have similar ones, so even a real dependence of  $C$  on that property would be blurred and difficult to learn. Our tests therefore concern the predictors as currently observed, not the bed properties they represent. In addition, the inferred  $C$  depends on the adopted stress balance, basal law, prescribed ice rheology, and regularized inversion, rather than representing a unique physical property of the bed. The ice softness is prescribed rather than inverted, so  $C$  also absorbs any error in it.

The point-wise formulation itself may also be limiting. The MLP uses only the predictor values at each location and cannot use information from neighboring locations. At the same time, membrane stresses couple the velocity response across space, so changes in  $C$  at one location can affect velocity elsewhere.

This is consistent with theoretical limits on local basal inference, in which basal disturbances are filtered through spatially coupled ice flow before appearing at the surface (Gudmundsson, 2003; Raymond and Gudmundsson, 2005; Gudmundsson and Raymond, 2008). The present experiments therefore identify several possible explanations for the remaining errors, but do not determine which mechanism is responsible in any individual region.

## CONCLUSION

This study tested whether inversion-derived basal-friction control  $C$  can be represented by a point-wise relationship with commonly available ice and subglacial observables. We evaluated this relationship in ten spatially held-out regions across the Amundsen Sea sector and in a separate PIG catchment test by inserting the predicted  $C$  fields into Icepack and comparing the resulting velocities with observations.

The tested local observables contain information that can produce  $C$  fields that improve modeled velocity relative to uniform  $C$  in held-out regions, but that relationship is not spatially consistent across the Amundsen Sea sector. The ice-product configurations give the lowest median velocity errors, and the combined configurations improve on a spatially uniform- $C$  baseline in nine of ten squares. However, performance varies strongly across locations; adding predictors does not systematically improve performance, and poor performance can occur even when the training data represent the held-out predictor values well. Training density provides no simple explanation either: lower training density does not consistently correspond to larger  $C$  errors across the square tests. Withholding all of PIG shows the same limitation at catchment scale, with CFG02 reducing velocity RMSE by only about 8% relative to uniform  $C$ . Although CFG02 lowers local velocity error over most of PIG, the remaining error is strongly concentrated in fast flow. The spatially withheld squares show that high predictor support can coexist with poor velocity performance. Agreement with the reference  $C$  is also only partly related to velocity performance: local  $C$  error and velocity error are positively associated, but not one-to-one, and over a region, closer overall agreement with  $C$  does not translate proportionally into lower velocity RMSE.

Under the framework considered here, the Amundsen Sea sector, the selected data products, a point-wise MLP, and a shallow-shelf/Weertman inversion, the tested ice and subglacial observables yield fitted controls that improve modeled velocity in most test regions, but they do not support a single point-wise relationship with inversion-derived  $C$  that is consistently successful across locations. More broadly, the results show that such parameterizations should be tested in spatially held-out regions and evaluated

through modeled velocity, not only through agreement with reference  $C$ . The same framework can be used to test other predictors and methods that use information from neighboring locations.

## CODE AND DATA AVAILABILITY

The inversion, velocity-simulation, and model-training code is available at <https://github.com/racheetmatai/icepack-inverse>. The reproduction artifacts, comprising the scientific inputs, the frozen experiment design, the canonical dataset and exact splits, the trained models, the friction-control fields and forward-simulation outputs, and the figure sources, are archived at Zenodo (<https://doi.org/10.5281/zenodo.22839669>). BedMachine Antarctica and the MEaSUREs velocity mosaic are not redistributed there; they remain available from NSIDC and are identified by exact checksums in the code repository.

## CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

## STATEMENT ON THE USE OF AI

We used generative AI tools to assist with language revision, consistency checking, and code refactoring. The authors reviewed all scientific decisions, analyses, interpretations, code, and the final text.

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## 725 ICE-FLOW EQUATIONS, GROUNDING TRANSITION, AND INVERSION

## 726 METHOD

Icepack solves the depth-integrated shallow-shelf stress balance

$$\nabla \cdot (h\mathbf{M}) + \boldsymbol{\tau}_b = \rho_i g h \nabla s, \quad (7)$$

where  $h$  is ice thickness,  $s$  is surface elevation,  $\rho_i$  is ice density,  $g$  is gravitational acceleration, and  $\boldsymbol{\tau}_b$  is the basal shear stress acting on the ice. For horizontal depth-averaged velocity  $\mathbf{u}$ , the membrane-stress tensor used here is

$$\mathbf{M} = 2\mu [\dot{\boldsymbol{\epsilon}} + \text{tr}(\dot{\boldsymbol{\epsilon}})\mathbf{I}], \quad \mu = \frac{1}{2} A^{-1/n} \dot{\epsilon}_e^{1/n-1}, \quad (8)$$

727 with  $\dot{\boldsymbol{\epsilon}} = (\nabla \mathbf{u} + \nabla \mathbf{u}^T)/2$ ,  $\dot{\epsilon}_e = [(\dot{\boldsymbol{\epsilon}} : \dot{\boldsymbol{\epsilon}} + \text{tr}(\dot{\boldsymbol{\epsilon}})^2 + \dot{\epsilon}_{\min}^2)/2]^{1/2}$ ,  $\mathbf{I}$  the horizontal identity tensor,  $\dot{\epsilon}_{\min} = 10^{-5} \text{ a}^{-1}$   
 728 Icepack’s default strain-rate floor,  $A$  the temperature-dependent rate factor, and  $n = 3$  the Glen exponent.  
 729 The modeled velocity is depth-averaged. Its comparison with the MEaSURES surface velocity therefore  
 730 uses the shallow-shelf assumption that vertical shear is negligible. We use the depth-averaged englacial  
 731 temperature of Van Liefferinge and Pattyn (2013) and linearly interpolate this field to the model mesh and  
 732 supply it in kelvin to Icepack’s rate-factor function to calculate  $A$  (Shapero and others, 2021). This differs  
 733 from the surface-air-temperature predictor used by the MLP.

The basal law is Eqn 2, with the Icepack sliding exponent  $m = 3$  in that equation. The dimensionless basal-friction control  $C$  is distinct from the dimensional basal-friction coefficient  $C_b$  that enters the friction

law given in equation 9.

$$\begin{aligned}
 p_w &= \rho_w g \max(0, h - s), & p_i &= \rho_i g h, \\
 \phi &= \begin{cases} 0, & p_i = 0, \\ \max[1 - (p_w/p_i)^p, 0], & p_i > 0, \end{cases} \\
 C_b &= \phi C_0 \exp(C).
 \end{aligned} \tag{9}$$

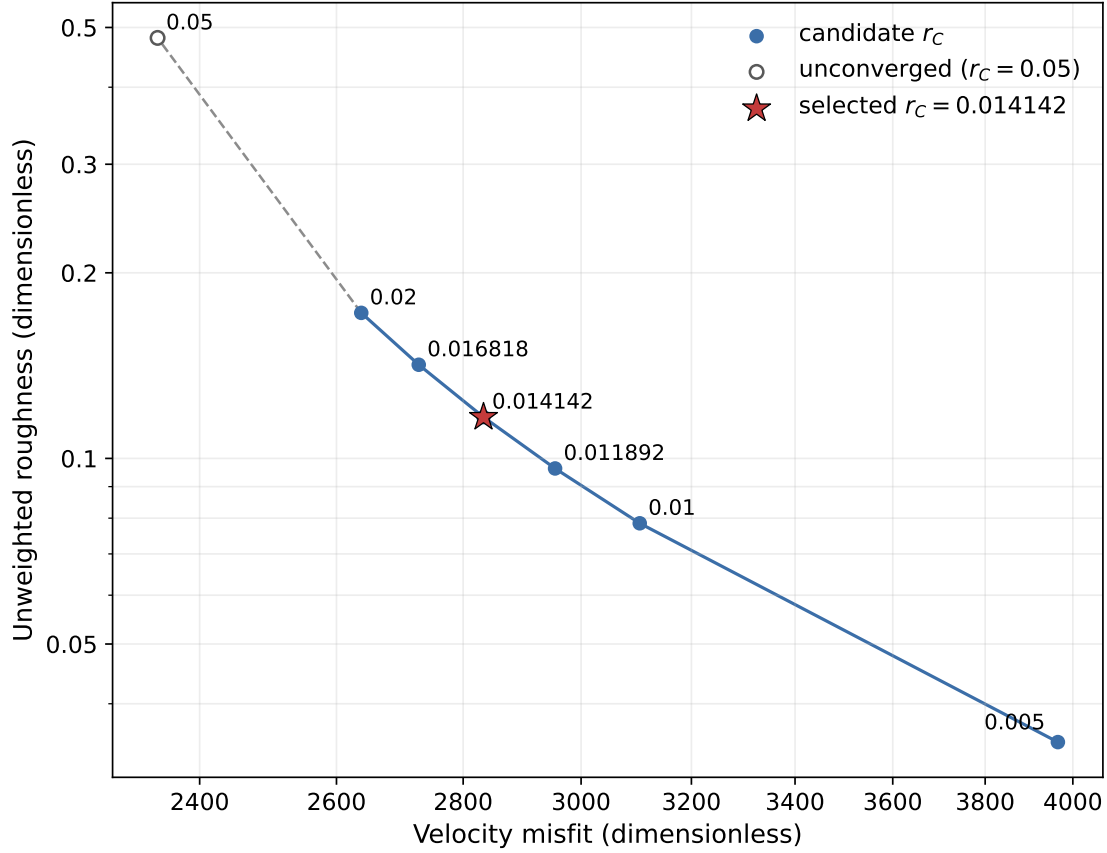
Here  $\rho_w$  is seawater density,  $p_w$  is the hydrostatic water-pressure estimate,  $p_i$  is ice overburden pressure,  $p = 1$  is the ramp exponent used in this study, and  $C_0 = 0.01 \text{ MPa (m a}^{-1}\text{)}^{-1/3}$ . Consequently  $C_b$  has units  $\text{MPa (m a}^{-1}\text{)}^{-1/3}$ , while  $C$  and  $\phi$  are dimensionless. The ramp is continuous and piecewise, with abrupt changes in slope introduced by the maximum operations.

Quadratic finite-element basis functions are used. We prescribe observed velocities along the non-terminus boundaries and apply an ocean-pressure traction condition at the termini. In that terminus condition the water-depth magnitude is  $d = \max(h - s, 0)$  and the normal-stress contribution is proportional to  $(\rho_i g h^2 - \rho_w g d^2)/2$ . At 11 edge points without a valid MEaSUREs value, we used the nearest valid pixel.

We initialized the inversion with  $C = 0$  and ran it for up to 300 optimization iterations. It converged after 257 iterations, when the gradient norm was  $6.7 \times 10^{-5}$ , below the  $10^{-3}$  acceptance threshold, and the misfit and roughness had stopped changing (relative changes below  $4 \times 10^{-5}$  over the final iterations). Figure 8 shows the L-curve used to select the regularization parameter. The maximum-curvature criterion selected  $r_C \simeq 0.014$  from the refined candidates. The adjacent candidate,  $r_C \simeq 0.017$ , had curvature within approximately 7% of the selected value, indicating a narrow elbow range rather than a uniquely resolved regularization value. We also tried values of  $r_C$  outside this range. With weaker smoothing,  $r_C = 0.05$  did not meet the convergence criteria within the iteration limit, and  $r_C = 0.1$  failed numerically.

## SPATIAL SEPARATION AND BUFFER CHOICE

For each 50 km evaluation square, we exclude an additional 40 km buffer from MLP training and validation on all sides, giving a 130 km excluded footprint. The buffer is intended to reduce the influence of nearby, spatially correlated training data on the withheld-square test. We chose the 40 km distance before model fitting using the spatial scale of observed ice-speed variations.



**Fig. 8.** L-curve for the sector-wide inversion. The axes are the two dimensionless terms of the objective in Eqn 4: the velocity misfit and the smoothness term without its factor  $r_C^{-2}$ . The maximum-curvature criterion selects  $r_C \simeq 0.014$ ; smaller  $r_C$  corresponds to stronger smoothing in the objective function in Eqn 4. The neighboring candidate near  $r_C \simeq 0.017$  has similar curvature, indicating a narrow elbow range.

Using MEaSUREs speed  $v$ , we formed

$$q(\mathbf{x}) = \ln[1 + v(\mathbf{x})/v_0], \quad v_0 = 1 \text{ m a}^{-1}. \quad (10)$$

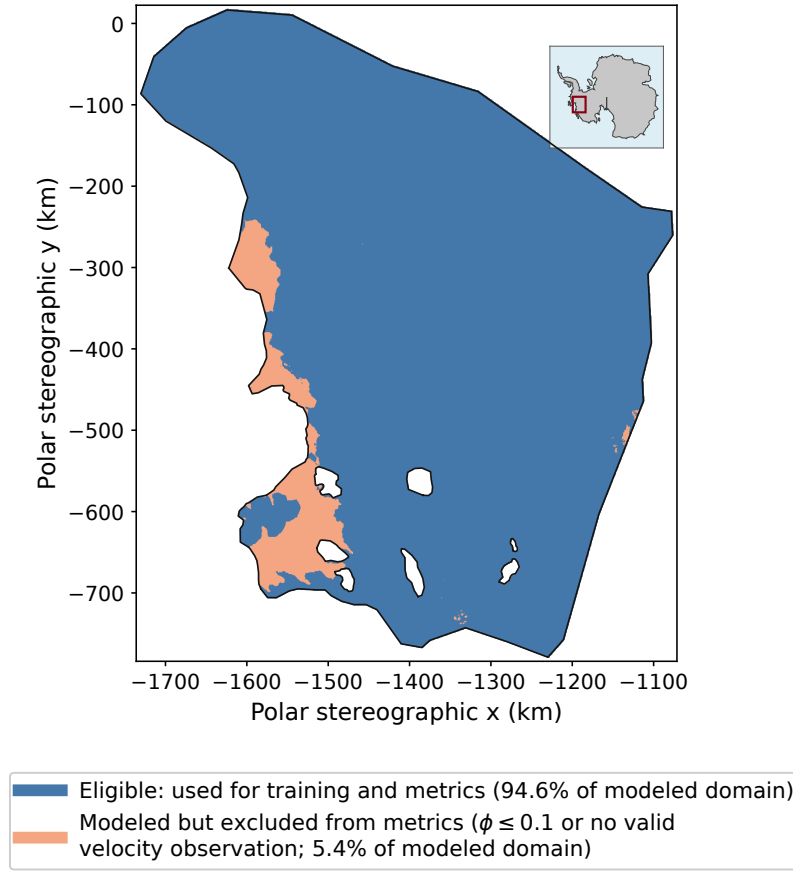
The logarithm reduces the influence of the fastest outlets, while the offset keeps the transformation finite where speed approaches zero. We then removed the broad sector-scale variation using a normalized Gaussian average with a 50 km standard deviation, leaving local variations in log speed. These local variations have an estimated  $1/e$  correlation length of 22.6 km, and their correlation falls to 0.108 at a separation of 40 km. Varying the Gaussian scale from 30 to 100 km yields  $1/e$  correlation lengths of 16.1–46.0 km.

Fast-flow widths that we measured from the MEaSUREs speed map (Mouginot and others, 2019) have a median of 10.5–18 km and a 90th percentile of 36–38 km across speed thresholds of 100, 250, and 500  $\text{m a}^{-1}$ , so the 50 km test side exceeds the width of most fast-flowing tributaries in the sector and spans ten cells of the coarsest 5 km predictors. Increasing the buffer from 40 to 50 km would reduce the local log-speed-anomaly correlation only from 0.108 to 0.072 while enlarging each excluded footprint from 16,900 to 22,500  $\text{km}^2$ , a 33% increase. We therefore selected 40 km to balance separation from training rows against the loss of training and validation data. Observed speed is used here as a proxy for the spatial scale of the flow, not of  $C$  itself. The buffer does not isolate the withheld square physically: the sector-wide inversion and the non-local membrane-stress response still couple it to the surrounding ice.

## ELIGIBLE ROWS, PREDICTOR CONSTRUCTION, AND SUPPORT

This section defines the common set of observation-grid rows used across all predictor configurations, describes how the derived predictors are constructed, and explains how predictor support is evaluated in the held-out regions. Of the MEaSUREs pixels with valid values of both velocity components, 1,622,598 are located in the model mesh by Firedrake’s point-location routine. All twelve predictors are finite at those rows. Applying  $\phi > 0.1$  leaves 1,530,992 common eligible rows, which are used by every predictor configuration. Figure 9 maps this eligible region against the complete modeled domain.

For the ten square experiments, excluding the 130 km footprint leaves 1,447,471–1,463,861 rows available for training and validation. Each MLP’s reproducible 90/10 split contains 1,302,723–1,317,474 training rows and 144,748–146,387 validation rows. The central-square evaluation sets contain 12,321 rows, except SQ03, which contains 12,432. For each training realization, every predictor configuration uses the same training locations and the same validation locations. Rows outside the common grounded eligible area



**Fig. 9.** The modeled domain and the eligible region used for training and for every reported metric. The excluded area (orange, 5.4% of the modeled domain) is where  $\phi \leq 0.1$  or no valid MEaSURES velocity is available. It is simulated but never scored; observations on floating ice still enter the inversion misfit. Every other map in the paper shows only the eligible region.

retain inversion-reference  $C$  in every forward simulation. White areas in maps lie outside the region shown for that experiment or outside the common mask; no locations were removed based on target values or model performance.

Surface elevation  $s$ , thickness  $h$ , and bed elevation  $b$  are represented as finite-element scalar fields. Their gradients are evaluated as finite-element vector expressions, interpolated component-wise to the observation grid, and summarized as  $\|\nabla f\| = (f_x^2 + f_y^2)^{1/2}$  for  $f \in \{s, h, b\}$ . Driving-stress magnitude is

$$\tau_d = \rho_i g h \|\nabla s\|, \quad (11)$$

reported in the stress unit used by Icepack, MPa.

Eqn 5 gives the alignment predictor. The small-gradient scales were estimated by dividing representative elevation-error scales by the median mesh-edge length of 6553.4 m. For the surface, a 1 m elevation-error

scale based on the reported accuracy of the Reference Elevation Model of Antarctica (Howat and others, 2019) gives  $\epsilon_s = 1.5 \times 10^{-4}$ . For the bed, the median BedMachine bed-elevation error over grounded mesh points is 35.31 m, giving  $\epsilon_b = 5.4 \times 10^{-3}$ . We rounded and fixed the resulting values before evaluating the model. We clip the alignment value to  $[-1, 1]$ . The  $\epsilon_b$  and  $\epsilon_s$  terms in the denominator reduce the magnitude of the alignment toward zero when either gradient is small relative to its corresponding scale. Table 1 summarizes the native spatial resolution and interpretation of the products used.

We evaluate marginal support separately for each predictor. A held-out point satisfies the marginal criterion when every predictor included in that configuration lies within the 1st–99th percentile range of the corresponding eligible non-held-out rows. Joint support evaluates the full combination of predictor values. We first sample the predictors on a 5 km grid covering the sector. A 5 km grid is used rather than the 450 m MEaSUREs grid because the predictors are finite-element fields on a mesh with a median edge length of 6.6 km and the coarsest products have 5 km spacing, so finer sampling would mainly add near-duplicate values; the grid also allowed the squares to be selected before the 450 m dataset was built. We transform each predictor to normal scores, replacing every value by the standard-normal quantile of its rank (Beasley and others, 2009), so that all predictors share a common, unitless scale. We then use principal component analysis (Jolliffe and Cadima, 2016) to reduce the predictor space while retaining at least 99% of the variance of the transformed predictors, and scale the retained components to unit variance. Each location is then a point in this transformed space, and the Euclidean distance between two points measures how different their predictor combinations are, with every retained component weighted equally. We use the same transformation for all spatial experiments.

A held-out point is jointly supported if some training location has a sufficiently similar predictor combination, that is, lies close to it in this space. To define how close is sufficient, we use the distances found within the sector itself. For each grid point, we calculate the distance in transformed predictor space to the most similar point located at least 40 km away along at least one grid axis, so that the excluded neighborhood is a square, matching the square shape of the 40 km buffer. The 95th percentile of these distances defines the joint-support threshold. For each held-out point, we then find the most similar 5 km reference-grid point within the corresponding training region. The held-out point satisfies the joint criterion when this distance does not exceed the threshold. This construction follows the area-of-applicability approach of Meyer and Pebesma (2021), which measures the distance from each prediction location to the most similar training location in a normalized predictor space.

We use the same construction twice: to select the squares, and to report support for each configuration. Square selection used all twelve predictors, so a candidate square had to be supported in the full predictor space: at least 95% of its area had to lie within the marginal bounds of each predictor, and at least 95% had to satisfy the joint criterion. The support values reported for each configuration use only that configuration’s predictors, transformation and threshold, so passing the selection check does not guarantee the same support for every configuration: a square selected with all twelve predictors can have lower support for an individual configuration (Fig. 5a). Reported predictor support is the area fraction satisfying both the marginal and joint criteria. We retain all held-out points in the modeled velocity evaluation, regardless of support.

## Training density and prediction error

Predictor support shows whether similar predictor combinations occur in the training data, but not how many there are. Here we test whether prediction error depends on that number, which we call the training density. For CFG02 and CFG04 in the ten central squares, and CFG02 in PIG, we measure the training density at each held-out row with  $d_{20}$ , the distance to its 20th-most-similar training row. We measure this distance in the transformed predictor space used for joint support, search all training rows, and apply no geographic separation. A small  $d_{20}$  means that many similar training rows exist; a large  $d_{20}$  means that few do. This is the standard nearest-neighbor measure of data density (Loftsgaarden and Quesenberry, 1965).

To judge whether a held-out  $d_{20}$  is small or large, we compare it with the  $d_{20}$  of 10,000 randomly chosen training rows, calculated in the same way. Its percentile  $p$  is the percentage of these training values that are smaller (ties count as half). We calculate  $p$  separately for each of the ten MLPs, whose training rows differ slightly, and use the median. A high  $p$  means that the held-out row has fewer similar training rows than a typical training row has. We group the rows into three categories,  $p \leq 50$ ,  $50 < p \leq 95$ , and  $p > 95$ , which we fixed, together with the neighbor count, before examining prediction errors. We do not rebalance the training data: the predictors are continuous and partly related, so balancing them one at a time would not balance their combinations. To test whether the neighbor count or training rows close by on the map affect the result, we repeat the analysis using the 100th and 500th most similar training rows, and using only training rows at least 40 km away.

Within each category, we calculate  $C$  RMSE between the median predicted  $C$  and the reference  $C$  at

the held-out rows, and velocity RMSE and relative RMSE over the same rows. Over whole squares, this  $C$  RMSE differs by at most 2.37% from the area-weighted values in Table 8.

For CFG02,  $C$  RMSE rises from the highest to the lowest density category in seven of the ten squares (Table 5), but the medians across the squares are almost identical (0.299, 0.297, and 0.301), and the median within-square correlation is only 0.069. CFG04 shows even less:  $C$  RMSE rises across the categories in only one square, and the median correlation is  $-0.052$ . Some squares show larger  $C$  errors where similar training rows are fewer, but training density does not generally explain  $C$  prediction error.

The result is the same within observed-speed classes. Below  $100 \text{ m a}^{-1}$ , the median within-square correlations are 0.080 for CFG02 and  $-0.113$  for CFG04; between 100 and  $500 \text{ m a}^{-1}$ , they are 0.079 and 0.015. Only one square contributes to the  $500\text{--}1000 \text{ m a}^{-1}$  class, too few to test faster flow. The result also does not depend on the neighbor count or on nearby training rows: using the 100th or 500th most similar training row, or only training rows at least 40 km away, gives median within-square correlations between  $-0.13$  and  $0.09$ .

Relative velocity RMSE does not follow training density either. In SQ06, CFG02 is worse than uniform  $C$  in the highest-density category (relative RMSE 1.15) but better in the lowest (0.96). In SQ05, its relative velocity RMSE rises from 0.37 in the highest-density category to 0.81 in the lowest, whereas for CFG04 in SQ02 it falls from 1.45 to 1.34. Lower training density therefore does not consistently mean larger velocity error.

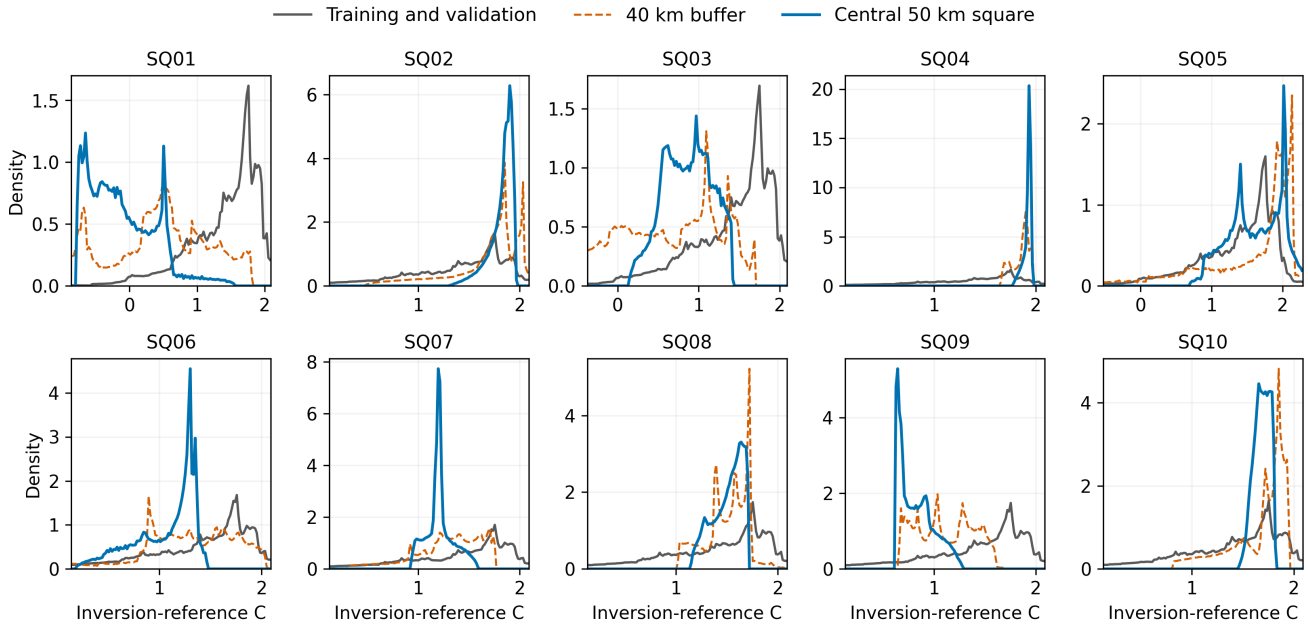
## BASAL-FRICTION CONTROL: DISTRIBUTIONS AND COMPARISONS

We first check whether the reference  $C$  values in each held-out region fall within the range of values available in the training-and-validation region. This is separate from predictor support, which considers only the input variables. We did not use reference  $C$  to select the square locations, define predictor support, or remove evaluation rows.

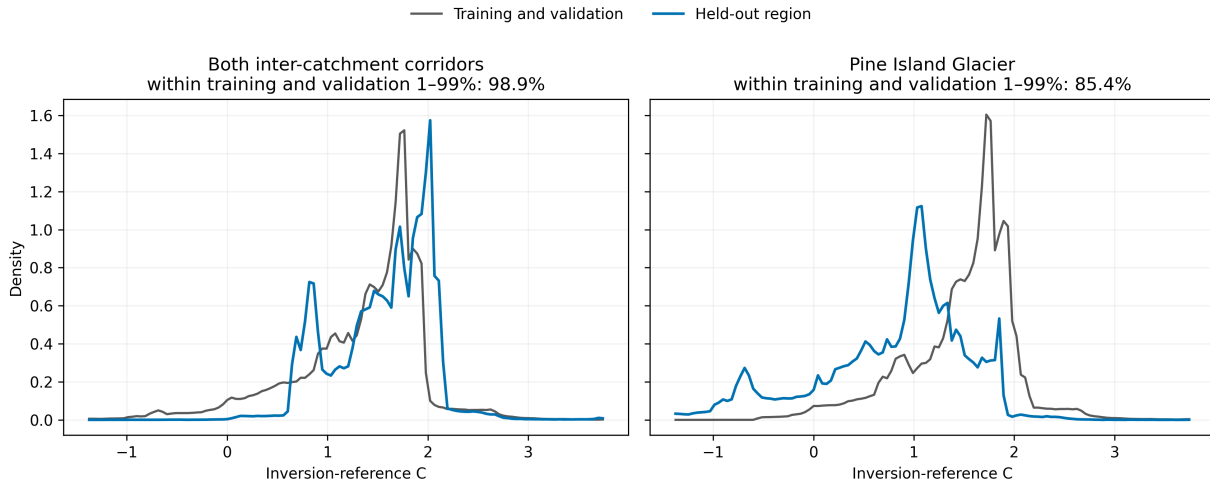
Figures 10, 11 and Table 6 compare the reference  $C$  distributions in the training-and-validation and held-out regions. Within each holdout experiment (one square, PIG, or the corridors), all configurations use the same training, validation, and held-out rows, so their reference- $C$  distributions are identical. These comparisons show how the reference- $C$  range in each held-out region overlaps that in the training-and-validation region; modeled velocity tests whether the predicted control reproduces the observed flow. The reference  $C$  is one of many possible regularized, model-dependent inversion solutions, and modeled velocity

**Table 5.**  $C$  RMSE of the median prediction against the inversion reference in the central squares, by training density. The columns give  $C$  RMSE for rows with  $p \leq 50$  (highest density),  $50 < p \leq 95$ , and  $p > 95$  (lowest density), where  $p$  is the  $d_{20}$  percentile. Spearman  $\rho$  is the within-square rank correlation between  $p$  and absolute  $C$  error; positive values mean larger errors where training density is lower. The last row for each configuration gives the median across the ten squares.

| Configuration | Square | $p \leq 50$ | $50 < p \leq 95$ | $p > 95$ | $\rho$ |
|---------------|--------|-------------|------------------|----------|--------|
| CFG02         | SQ01   | 0.371       | 0.410            | 0.418    | 0.076  |
|               | SQ02   | 0.206       | 0.232            | 0.265    | 0.107  |
|               | SQ03   | 0.370       | 0.384            | 0.411    | 0.086  |
|               | SQ04   | 0.201       | 0.204            | 0.211    | 0.056  |
|               | SQ05   | 0.416       | 0.351            | 0.337    | -0.141 |
|               | SQ06   | 0.493       | 0.516            | 0.543    | 0.035  |
|               | SQ07   | 0.227       | 0.242            | 0.245    | 0.090  |
|               | SQ08   | 0.175       | 0.212            | 0.198    | 0.133  |
|               | SQ09   | 0.731       | 0.735            | 0.753    | 0.062  |
|               | SQ10   | 0.130       | 0.103            | 0.110    | -0.087 |
|               | Median | 0.299       | 0.297            | 0.301    | 0.069  |
| CFG04         | SQ01   | 1.100       | 1.298            | 1.058    | -0.101 |
|               | SQ02   | 0.608       | 0.634            | 0.470    | -0.049 |
|               | SQ03   | 0.512       | 0.572            | 0.587    | 0.139  |
|               | SQ04   | 0.375       | 0.364            | 0.236    | -0.637 |
|               | SQ05   | 0.473       | 0.508            | 0.422    | 0.065  |
|               | SQ06   | 0.383       | 0.515            | 0.474    | 0.224  |
|               | SQ07   | 0.634       | 0.477            | 0.466    | -0.257 |
|               | SQ08   | 0.348       | 0.307            | 0.309    | -0.044 |
|               | SQ09   | 0.517       | 0.491            | 0.411    | -0.055 |
|               | SQ10   | 0.299       | 0.317            | 0.302    | -0.094 |
|               | Median | 0.492       | 0.499            | 0.444    | -0.052 |



**Fig. 10.** Reference- $C$  distributions for the ten withheld squares. Each panel compares the training and validation rows with the central 50 km evaluation square and the surrounding buffer. We choose the horizontal range separately for each panel to show the central part of the distributions clearly. Each density is normalized using all rows in its set, including values outside the displayed range. Numerical 1st–99th-percentile intervals and the percentage of held-out values within the training-and-validation interval are reported in Table 6. Reference  $C$  was not used to select square locations or remove evaluation rows.



**Fig. 11.** Reference- $C$  distributions for the two regional tests. Each panel compares the training and validation rows with the complete held-out region. The percentage shown is the fraction of held-out  $C$  values that fall within the 1st–99th-percentile range of the training and validation data.

**Table 6.** Reference- $C$  distributions in the training-and-validation and held-out regions. Intervals are the 1st–99th percentiles of dimensionless  $C$ . The final column gives the percentage of held-out rows whose  $C$  values fall within the corresponding training-and-validation interval.

| Experiment      | Training and<br>validation rows | Held-out rows | Training and<br>validation interval | Held-out interval | Held-out within<br>training range (%) |
|-----------------|---------------------------------|---------------|-------------------------------------|-------------------|---------------------------------------|
| SQ01            | 1,448,295                       | 12,321        | −0.22–2.67                          | −0.76–1.30        | 51.9                                  |
| SQ02            | 1,447,471                       | 12,321        | −0.69–2.67                          | 1.43–1.96         | 100.0                                 |
| SQ03            | 1,450,403                       | 12,432        | −0.69–2.67                          | 0.21–1.40         | 100.0                                 |
| SQ04            | 1,447,471                       | 12,321        | −0.69–2.67                          | 1.80–1.96         | 100.0                                 |
| SQ05            | 1,459,429                       | 12,321        | −0.65–2.66                          | 0.84–2.35         | 100.0                                 |
| SQ06            | 1,447,471                       | 12,321        | −0.69–2.67                          | 0.28–1.43         | 100.0                                 |
| SQ07            | 1,447,471                       | 12,321        | −0.69–2.67                          | 0.95–1.55         | 100.0                                 |
| SQ08            | 1,453,825                       | 12,321        | −0.69–2.66                          | 1.18–1.71         | 100.0                                 |
| SQ09            | 1,458,393                       | 12,321        | −0.69–2.66                          | 0.62–1.23         | 100.0                                 |
| SQ10            | 1,463,861                       | 12,321        | −0.69–2.66                          | 1.50–1.81         | 100.0                                 |
| Inter-catchment | 1,241,683                       | 289,309       | −0.72–2.65                          | 0.49–2.66         | 98.9                                  |
| PIG             | 1,303,307                       | 227,685       | −0.08–2.69                          | −1.28–2.05        | 85.4                                  |

remains the primary evaluation in this study.

We measure agreement with the inversion-reference control using  $C$  RMSE and the coefficient of determination  $R_C^2$ . The latter is one minus the sum of squared prediction errors divided by the sum of squared deviations of reference  $C$  from its mean over the same evaluated locations. These diagnostics compare the reference  $C$  with the median of the ten MLP predictions at each location. The control used in the simulations equals this prediction except in mesh cells that cross the edge of the replaced region, where it blends into the reference. These cells contain none of the central-square rows; for PIG and for the full 130 km withheld areas around the squares, computing  $C$  RMSE from the simulated control instead of the median prediction changes it by at most 0.004.

Table 7 reports agreement with the inversion-reference control on the validation rows. Each MLP was evaluated on its own validation rows, so the table reports the median of the ten member-level results for each square-configuration experiment. In every square, CFG04 shows the weakest agreement with the reference  $C$  on its validation rows, with  $C$  RMSE of 0.36–0.40 and  $R_C^2$  of 0.65–0.70; the other five configurations have  $C$  RMSE of 0.02–0.08 and  $R_C^2$  of at least 0.98.

We also compare the predicted  $C$  fields directly with the reference  $C$  to measure how closely the MLP reproduces the inversion result. Table 8 reports  $C$  RMSE and  $R_C^2$  for every central square. We retain

**Table 7.** Agreement with the inversion-reference control on the validation rows. For each square-configuration experiment,  $C$  RMSE and  $R_C^2$  are computed separately for each of the ten MLPs on its own validation rows, and the entry is the median of the ten values. Each MLP has different validation rows, so the metrics are computed per MLP rather than from the median prediction. The validation rows lie outside the 130 km footprint and were used to choose each MLP’s stopping epoch, so they are not independent test data. Both quantities are dimensionless.

| Square | CFG01       | CFG02       | CFG03       | CFG04       | CFG05       | CFG06       |
|--------|-------------|-------------|-------------|-------------|-------------|-------------|
| SQ01   | 0.077/0.984 | 0.075/0.985 | 0.056/0.992 | 0.363/0.652 | 0.020/0.999 | 0.022/0.999 |
| SQ02   | 0.078/0.987 | 0.081/0.986 | 0.057/0.993 | 0.387/0.680 | 0.023/0.999 | 0.022/0.999 |
| SQ03   | 0.072/0.988 | 0.077/0.986 | 0.070/0.989 | 0.383/0.661 | 0.022/0.999 | 0.021/0.999 |
| SQ04   | 0.080/0.986 | 0.079/0.986 | 0.056/0.993 | 0.393/0.666 | 0.023/0.999 | 0.022/0.999 |
| SQ05   | 0.076/0.987 | 0.077/0.986 | 0.055/0.993 | 0.364/0.697 | 0.023/0.999 | 0.021/0.999 |
| SQ06   | 0.073/0.988 | 0.080/0.986 | 0.057/0.993 | 0.382/0.681 | 0.022/0.999 | 0.022/0.999 |
| SQ07   | 0.077/0.987 | 0.077/0.987 | 0.058/0.993 | 0.388/0.681 | 0.021/0.999 | 0.022/0.999 |
| SQ08   | 0.077/0.988 | 0.079/0.987 | 0.056/0.993 | 0.397/0.669 | 0.022/0.999 | 0.023/0.999 |
| SQ09   | 0.077/0.987 | 0.079/0.986 | 0.057/0.993 | 0.384/0.676 | 0.022/0.999 | 0.022/0.999 |
| SQ10   | 0.077/0.987 | 0.080/0.986 | 0.058/0.993 | 0.391/0.676 | 0.023/0.999 | 0.022/0.999 |

negative  $R_C^2$  values; a negative value means that the predicted  $C$  has larger squared error than using the mean reference  $C$  within that square. The median  $R_C^2$  values across the ten squares are  $-1.39$ ,  $-1.35$ ,  $-7.97$ ,  $-6.52$ ,  $-1.41$ , and  $-1.21$  for CFG01–CFG06, respectively. Poor agreement with the reference  $C$  can still coexist with improved modeled velocity, so these statistics answer a different question from the velocity comparison.

For the PIG holdout experiment, the median validation  $C$  RMSE /  $R_C^2$  across the ten MLP members is  $0.081/0.981$  for CFG01,  $0.077/0.983$  for CFG02, and  $0.056/0.991$  for CFG03. These values were calculated on each member’s validation rows outside the withheld PIG region. Despite this strong validation agreement, the corresponding median predicted fields have PIG holdout  $C$  RMSE /  $R_C^2$  values of  $0.92/-0.49$ ,  $0.85/-0.28$ , and  $1.23/-1.69$ , respectively. Thus, performance on randomly withheld validation rows does not predict spatial transfer to the withheld catchment.

The inter-catchment experiment withholds both corridors, PIG–Thwaites and Thwaites–Dotson, and trains on the three named catchments. It tests whether a relationship learned within the catchments transfers to the regions between them. We trained only CFG04–CFG06 for this experiment, so it is not directly comparable with the six-configuration square tests. As for the squares and PIG, the three predicted controls and the uniform control replace the inversion reference only inside the corridors and keep the reference  $C$  outside them. The median validation  $C$  RMSE /  $R_C^2$  across the ten MLP members is

**Table 8.** Comparison between predicted and reference  $C$  in the ten central squares. Each entry gives  $C$  RMSE /  $R_C^2$  for the median predicted  $C$  field, computed on finite-element cells and weighted by cell area. RMSE is in dimensionless  $C$  units. Values quoted on the 450 m observation rows elsewhere in the text differ by at most 0.01.

| Square | CFG01       | CFG02       | CFG03       | CFG04       | CFG05       | CFG06       |
|--------|-------------|-------------|-------------|-------------|-------------|-------------|
| SQ01   | 0.42/0.25   | 0.41/0.29   | 1.44/−8.00  | 1.21/−5.42  | 0.58/−0.47  | 0.52/−0.20  |
| SQ02   | 0.22/−2.69  | 0.22/−2.46  | 0.88/−56.51 | 0.61/−26.00 | 0.17/−1.04  | 0.15/−0.69  |
| SQ03   | 0.38/−0.63  | 0.37/−0.61  | 0.54/−2.31  | 0.54/−2.32  | 0.35/−0.38  | 0.33/−0.29  |
| SQ04   | 0.20/−29.81 | 0.20/−30.02 | 0.36/−98.61 | 0.33/−80.67 | 0.16/−18.76 | 0.16/−18.71 |
| SQ05   | 0.37/0.03   | 0.37/0.02   | 0.46/−0.50  | 0.48/−0.63  | 0.41/−0.21  | 0.42/−0.25  |
| SQ06   | 0.52/−1.83  | 0.51/−1.72  | 0.51/−1.77  | 0.47/−1.35  | 0.33/−0.14  | 0.34/−0.19  |
| SQ07   | 0.24/−2.44  | 0.23/−2.18  | 0.57/−17.72 | 0.52/−14.99 | 0.30/−4.13  | 0.31/−4.77  |
| SQ08   | 0.20/−0.96  | 0.20/−0.98  | 0.28/−2.78  | 0.31/−3.43  | 0.24/−1.77  | 0.24/−1.73  |
| SQ09   | 0.74/−18.37 | 0.74/−18.49 | 0.87/−26.27 | 0.49/−7.63  | 0.53/−8.90  | 0.51/−8.38  |
| SQ10   | 0.10/−0.79  | 0.10/−0.83  | 0.23/−7.93  | 0.31/−14.95 | 0.31/−15.21 | 0.27/−11.59 |

0.317/0.764 for CFG04, 0.021/0.999 for CFG05, and 0.020/0.999 for CFG06. Over both corridors together, the corresponding median predicted fields have holdout  $C$  RMSE /  $R_C^2$  values of 0.81/−1.86, 0.33/0.53, and 0.33/0.52. Their velocity RMSEs are 470.3, 119.0, and 109.3 m a<sup>−1</sup> (relative RMSE 0.81, 0.21, and 0.19), against 578.4 m a<sup>−1</sup> for uniform  $C$  and 28.1 m a<sup>−1</sup> for the inversion reference. All three configurations improve on uniform  $C$ , and CFG05 and CFG06 do much better than CFG04.

These are more favorable holdout conditions compared to PIG holdout: 98.9% of the held-out corridor  $C$  values fall within the training-and-validation 1st–99th-percentile interval, compared with 85.4% for PIG (Table 6), and the withheld corridors are bounded on both sides by catchments that remain in the training and validation dataset. The corridor values therefore indicate what the point-wise mapping achieves under favorable conditions rather than general transferability, and even there the best predicted controls, CFG05 and CFG06, leave velocity RMSE about four times that of the inversion reference.

## MLP TRAINING

This section describes the MLP architecture and training procedure and the ten training runs for each experiment.

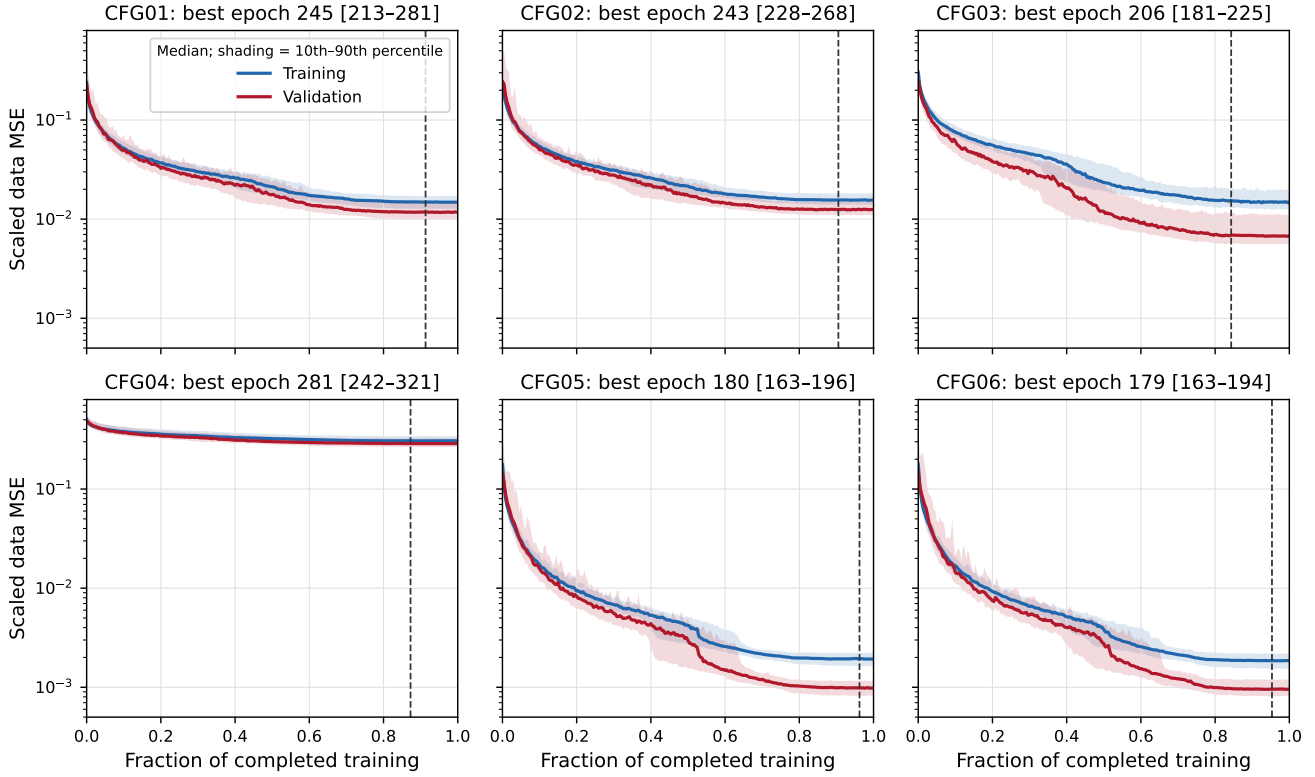
The MLP has ten hidden layers of 200 units. Each hidden layer is followed by batch normalization and a sigmoid linear unit (SiLU) activation, and the final layer is linear. Predictors and  $C$  are robustly scaled

using medians and interquartile ranges fitted on training rows only. Adam uses an initial learning rate of  $10^{-3}$  and batch size 1024, with reproducible epoch-wise shuffling. The learning rate is multiplied by 0.2, down to  $10^{-6}$ , after 15 epochs without a decrease in validation mean squared error (MSE) of at least  $10^{-4}$ . Training stops after 45 epochs without such a decrease, and we keep the weights from the epoch with the lowest validation MSE.

A separate CFG06 calculation selected the L2 penalty without using any of the ten central squares. It retained the square buffers, PIG and the inter-catchment corridors, so the regional tests were withheld from their final MLP fits but not from this calibration. Its fixed 70/20/10 split contained 1,407,671 rows: 985,369 for training, 281,534 for validation, and 140,768 for testing. We applied L2 penalties of 0,  $10^{-6}$ ,  $10^{-5}$ , and  $10^{-4}$  to the hidden-layer weights and compared them using unpenalized validation MSE. The respective minimum values were 0.001207, 0.001332, 0.001583, and 0.003117. No nonzero penalty reduced validation MSE below that of the zero-penalty model, so all final MLP trainings used  $\lambda_{L2} = 0$ .

We did not optimize the network architecture separately for each predictor configuration or held-out region. Instead, we used the same relatively large network, with ten hidden layers of 200 units, for all experiments. This architecture provides substantial flexibility to represent nonlinear relationships between the predictors and  $C$ , while keeping the network design fixed across configurations and spatial tests. We limit overfitting through a separate validation set, learning-rate reduction, early stopping, and restoring the model with the lowest validation MSE. The training and validation histories (Fig. 12) show whether training converged and whether the networks overfit. A full architecture search would require separate tuning across multiple predictor configurations and spatial splits and was beyond this study's scope. The chosen architecture should therefore be viewed as a fixed, flexible modeling choice rather than an optimized architecture.

The complete histories in Fig. 12 show that the training and validation losses decrease and then approach a plateau. Median validation MSE decreases from its initial value by factors of 19.5, 19.4, 37.7, 1.68, 148, and 162 for CFG01–CFG06, respectively. After the early epochs, median validation MSE stays below median training MSE, ending at about half of it for CFG03, CFG05, and CFG06. Training MSE is higher because it is computed during each epoch, with batch normalization using the statistics of each batch rather than the running averages used for validation. The histories therefore provide no clear evidence that the relatively large network is increasingly fitting the training rows at the expense of the validation rows. Validation agreement with the reference  $C$  is high for CFG01, CFG02, CFG03, CFG05, and CFG06, while



**Fig. 12.** Training and validation loss during training for the 600 square MLPs. Each panel shows one configuration: ten squares with ten MLPs each, 100 runs in total. Runs stop at different epochs, so the horizontal axis gives each run’s progress from its first epoch (0) to its last (1). Lines show the median training and validation MSE of the 100 runs, and shading shows their 10th–90th percentile range. MSE is computed on robustly scaled  $C$  and measures the fit to the reference  $C$ , not modeled velocity. Training MSE lies above validation MSE after the first epochs because of how it is computed (see text). The dashed line marks the median position of the epoch whose weights were kept, the one with the lowest validation MSE; the panel title gives that epoch’s median number and, in brackets, its interquartile range. No held-out square was used to choose the kept weights.

CFG04 shows a much smaller reduction in validation loss and higher validation  $C$  RMSE and lower  $R_C^2$ . However, CFG04 still reduces velocity RMSE relative to uniform  $C$  in seven of the ten spatially withheld squares. The main implication is that fitting  $C$  well on random validation rows within the training region does not, by itself, predict performance in a spatially separated region. The common architecture and training procedure provide a controlled comparison among predictor configurations, while the spatially withheld tests assess whether the learned relationships produce useful modeled velocities elsewhere.

Each combination of holdout test and predictor configuration (for example, SQ03 with CFG02) uses ten reproducible training runs, which differ in their training/validation row assignments, initial weights, and epoch shuffles. The ten squares use all six configurations. The two regional tests use three configurations each, chosen before training by how well the withheld region’s predictor values are covered by the training data (the marginal and joint support described in Appendix section Eligible rows, predictor construction,

**Table 9.** Spearman correlations between reported BedMachine bed-elevation error and local velocity error in the ten central squares. Each configuration is summarized by the median and range of its ten within-square correlations.

| Configuration | Median | Minimum | Maximum |
|---------------|--------|---------|---------|
| CFG01         | −0.005 | −0.117  | 0.139   |
| CFG02         | −0.018 | −0.135  | 0.137   |
| CFG03         | 0.012  | −0.151  | 0.082   |
| CFG04         | 0.041  | −0.139  | 0.138   |
| CFG05         | −0.062 | −0.218  | 0.193   |
| CFG06         | −0.059 | −0.214  | 0.181   |

and support): CFG01–CFG03 for PIG and CFG04–CFG06 for the inter-catchment corridors, so that each regional test uses the configurations least likely to fail simply because its predictor values lie outside the training range. This gives 66 combinations and 660 MLPs in total. For each combination, we take the median of the ten predicted  $C$  fields at each control point. The ten individual runs show how sensitive the predictions are to the training procedure and are combined into the median; they are not independent spatial tests or estimates of predictive uncertainty.

## BEDMACHINE RECONSTRUCTION AND VELOCITY ERROR

Several predictors are derived from BedMachine, so we also examined whether differences in modeled velocity performance could be related to the BedMachine reconstruction itself. We considered two quantities: the source class, which identifies the method used to construct the bed elevation, and the reported bed-elevation error. Our aim is to test the hypothesis in Eqn 1 rather than to build the best predictor of  $C$ , so we used these quantities only to examine the results, not as MLP predictors or to select evaluation rows. The analysis uses the ten central 50 km squares and the velocity from each configuration’s median- $C$  forward simulation.

For each square and configuration, we calculated the Spearman correlation, across the MEaSURES grid cells in the square, between reported bed-elevation error and local velocity-vector error,  $\|\mathbf{u}_{\text{ML}} - \mathbf{u}_{\text{obs}}\|$ . Table 9 summarizes the ten within-square correlations. Their small magnitudes and changing signs show that larger reported bed-elevation errors did not consistently coincide with larger velocity errors.

BedMachine source class records how the geometry supplied to the MLP was reconstructed. Class 5 denotes streamline-diffusion interpolation, which smooths thickness preferentially along flow; it is not

a direct bed measurement. Eight central squares contain only class 5 rows. SQ05 contains 838 mass-conservation rows (class 2), 174 interpolation rows (class 3), and 11,309 class 5 rows; SQ06 contains 12, 18, and 12,291 rows in these classes, respectively. These are counts of evaluated observation-grid rows.

Reconstruction method changes little within most squares, so these tests cannot separate its effect from other differences between locations. Performance also varies among squares using the same method, so differences between source classes cannot alone account for the observed spatial variation.

## PREDICTING WHERE TRANSFER SUCCEEDS

The preceding sections examine, one at a time, how predictor support, training density, agreement with the reference  $C$ , and the BedMachine reconstruction relate to velocity performance. Here we test whether the predictors, taken together, can show in advance where the predicted  $C$  improves modeled velocity compared with uniform  $C$ .

For each observation-grid row (one MEaSUREs 450 m grid cell) in a withheld region, we recorded two outcomes. The first, the improvement criterion, is whether the predicted  $C$  gives a lower local velocity error than uniform  $C$ ,  $e_{\text{ML}} < e_{\text{uniform}}$ , where  $e$  is the local vector-error magnitude defined in the Methods. The second, the halving criterion,  $e_{\text{ML}} < 0.5 e_{\text{uniform}}$ , requires the predicted  $C$  to at least halve the local error; it checks that the results do not depend on counting small improvements as success.

We then trained random forest classifiers (Breiman, 2001) to predict each outcome from local inputs; a random forest can capture nonlinear relationships among many inputs without specifying their form. The classifiers predict the outcome rather than  $C$ , so observed speed, which is excluded from the MLP predictors, can be used as an input. We compared three sets of inputs, each with its own classifier: observed speed alone, the twelve predictors, and the twelve predictors together with observed speed. These input sets are the same for every configuration, so they include all twelve predictors even when the MLP used only some of them. All classifiers use the same settings: 120 trees, each at most 14 splits deep, at least 40 rows in each final group, and the two outcomes weighted equally.

We tested each classifier on a region it had not seen. For the squares, we trained on the central squares of nine tests and tested on the central square of the tenth, repeating this for every square and configuration. Each training row therefore comes from a predicted  $C$  field produced without that location. For PIG, we trained on the central squares of all ten tests and tested CFG02 over the whole catchment. We left out the 18,409 training rows that lie inside PIG, so the classifier never sees the tested area.

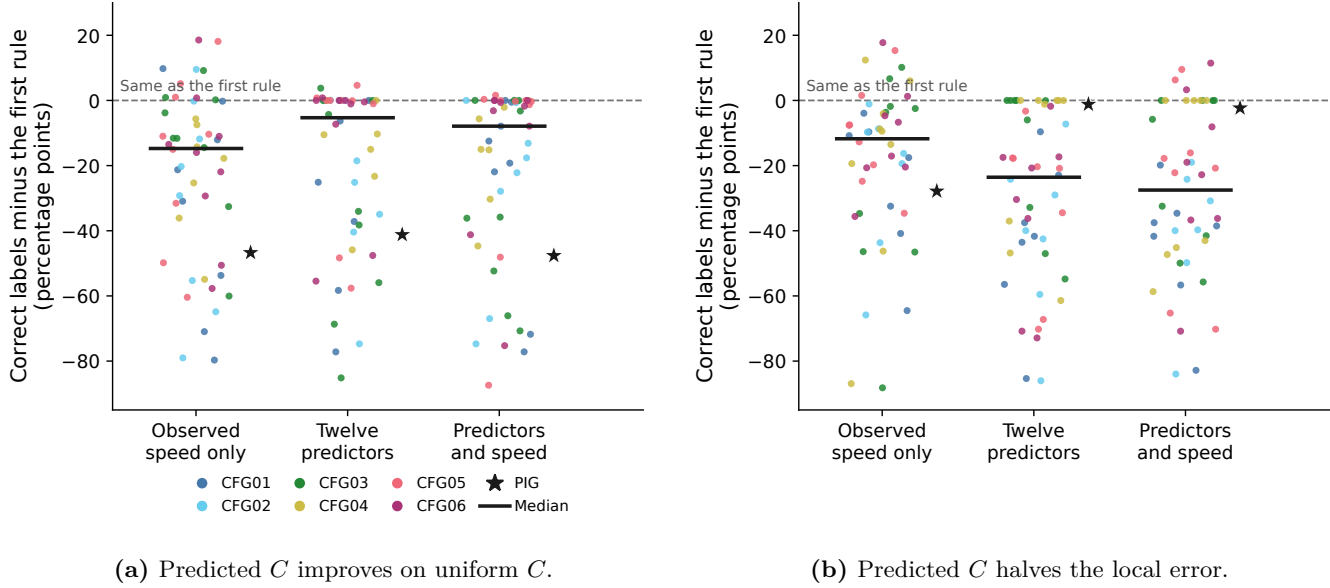
We report the percentage of test rows that each classifier labels correctly and compare it with two simple rules that give the same answer at every row of the tested region, a standard reference for classification accuracy known as the no-information rate (Kuhn and Johnson, 2013, Chapter 11). The first rule gives the outcome that is more common in the tested region itself: if the predicted  $C$  improves on uniform  $C$  at 70% of its rows, it answers “improves” everywhere and is 70% correct. It requires the test outcomes, which are not known in advance. The second rule gives the outcome that is more common in the training squares, whatever happens in the tested region; like the classifiers, it uses only the training data. To show that it can identify where the predicted  $C$  succeeds within a region, a classifier must label a higher percentage of rows correctly than both rules.

The ten squares, six MLP configurations, three classifier input sets, and two criteria (improvement and halving) give 360 tests. In some squares the outcome was the same at every row, so there was nothing for a classifier to separate. This happened for 10 combinations of square and configuration under the improvement criterion and 12 under the halving criterion; in SQ04, for example, the predicted  $C$  improved on uniform  $C$  at every row for all six configurations. Leaving out these combinations, with their three input sets, removes 66 tests and leaves 294. Of the 60 combinations of square and configuration, 50 remain under the improvement criterion and 48 under the halving criterion.

Across these 294 tests, the classifiers labeled 1,967,017 of 3,626,370 test rows correctly (54.2%), and the median score of a test was 51.4%. The first rule, which gives the tested square’s more common outcome at every row, had a median of 77.5%, and the classifiers did better than it in only 32 of the 294 tests. The second rule, which gives the training squares’ more common outcome, had a median of 70.0%, and the classifiers did better than it in 98 of the 294 tests.

The two criteria and three input sets, taken separately, show the same result. For the improvement criterion, the median percentages correct were 57.9 with observed speed alone, 62.6 with the twelve predictors, and 61.6 with both, against 77.6 for the first rule (Fig. 13a). For the halving criterion they were 59.2, 39.6 and 39.1, against 74.3 (Fig. 13b). Adding the twelve predictors to observed speed did not help. Using both, the classifier did better than the first rule in 5 of the 50 combinations under the improvement criterion and in 4 of the 48 under the halving criterion; using observed speed alone, it did so in 10 of 50 and 8 of 48. Against the second rule, the classifier using both did better in 14 of 50 and 19 of 48 combinations, and the one using observed speed alone in 17 of 50 and 22 of 48.

These failures are not a matter of the classifier fitting its training data poorly. The same classifier



**Fig. 13.** Ability of a random forest classifier to predict where the predicted  $C$  improves on uniform  $C$ . Each point is one withheld square and configuration, and gives the percentage of held-out rows labeled correctly minus the percentage obtained by predicting whichever outcome is more common in that square; color identifies the configuration. Values below the dashed line are no better than that rule. The classifier is trained on the central squares of the other nine tests. Stars show the PIG CFG02 test, for which the classifier is trained on all ten central squares with rows inside PIG removed. Horizontal bars mark medians across the square tests. Panel (a) uses  $e_{ML} < e_{uniform}$  and panel (b) uses  $e_{ML} < 0.5 e_{uniform}$ . We omit tests in which every row or no row met the criterion.

labeled 98.7% of its training rows correctly, and 98.3% of rows held out at random from within the training squares, compared with about half of the rows in a spatially separate square. Nor do they depend on where the classifier's output is cut into a yes or no decision. Choosing that cut-off on rows held out from the training squares, using only training data, raised the median to 54.1% and did better than the first rule in 29 of the 294 tests. Even choosing the best possible cut-off for each withheld square after seeing its outcomes raised the median to 81.6% and did better than the first rule in only 122 of the 294 tests.

The classifiers also failed to transfer to PIG, where 227,685 rows were evaluated. For the improvement criterion they labeled 30.8%, 36.4% and 29.9% of rows correctly for the three input sets, against 77.6% for the first rule. For the halving criterion they labeled 64.2%, 90.8% and 89.7% correctly, against 92.0%. No input set did better than the first rule in either case.

Part of the reason is that the relationship between ice speed and success is reversed between the two settings. In the rows used to train the PIG classifier, the predicted  $C$  meets the improvement criterion at 98.5% of rows in the slowest quarter of speeds but at only 48.3% and 52.1% in the two fastest quarters. In PIG, it is met at 51.3% of rows in the slowest quarter of PIG speeds and 93.8% in the fastest. A classifier that learns where the predicted  $C$  succeeds in the squares therefore gets the pattern backwards in PIG.

The predictors therefore do not provide a reliable way to identify in advance where the predicted  $C$  will improve modeled velocity. The combined classifier, which uses observed speed and the twelve predictors, is not consistently better than the one using observed speed alone, rarely does better than the first rule, falls short of the second rule in most tests, and does not transfer from the squares to PIG. The same classifier labels its training squares almost perfectly, so the difficulty lies in transferring to a new region, not in the classifier’s ability to fit the data. The tests in which every row had the same outcome point the same way: whether the point-wise relationship succeeds depends mainly on the withheld region as a whole rather than on individual locations within it. Together with the tests of predictor support and training density, this indicates that the local predictor values, individually or together, do not determine where the predicted  $C$  succeeds. If  $C$  were a point-wise function of these predictors (Eqn 1), failures would be tied to particular predictor values; instead, they depend on the region, which does not support the point-wise hypothesis.

## SENSITIVITY TO THE REPLACEMENT REGION

For each square, predicted and uniform  $C$  are applied only within the complete 130 km excluded footprint where  $\phi > 0.1$ . The PIG experiment uses the corresponding catchment mask. Elsewhere, including across the grounding transition, we retain the inversion-derived reference  $C$ . This keeps the geometry, boundary conditions, and grounding treatment identical while holding the surrounding reference control fixed.

We also compared the controlled design with an earlier calculation in which the predicted  $C$ , or the corresponding uniform- $C$  value, replaced the inversion-reference control throughout the common eligible grounded sector. In that earlier design, the predicted control was therefore also used in the MLP training geography. In the controlled design used for the main results, replacement is restricted to the complete 130 km footprint for each square, or to the held-out regional geography, intersected with the same common eligible grounded mask. Predicted and uniform controls use identical masks, and both retain inversion-reference  $C$  outside the mask. The complete 40 km buffer around each central square is part of the replacement region.

We evaluated the effects of correcting the observational reference and changing the replacement geography separately. Replacing the mesh-interpolated observational values with the original paired MEaSUREs components changed the central-square MLP RMSE by a median of  $0.07 \text{ m a}^{-1}$  and a maximum of  $0.95 \text{ m a}^{-1}$ . Restricting replacement to the withheld geography then changed the central-square MLP RMSE by a median of  $0.003 \text{ m a}^{-1}$  and a maximum of  $2.02 \text{ m a}^{-1}$ . Neither change altered whether a con-

1088 figuration improved on uniform  $C$  in any square or which configuration had the lowest RMSE in a square.  
1089 Differences were larger over the complete footprints. For CFG02 and CFG04, the median relative RMSE  
1090 changed from 0.54 and 0.82 under sector-wide replacement to 0.64 and 0.86 under controlled replacement;  
1091 both configurations improved on uniform  $C$  in nine of ten complete footprints in both designs. Across  
1092 these 20 footprint comparisons, the median absolute change in relative RMSE was 0.04 and the largest was  
1093 0.28. The principal central-square findings are therefore robust to the two replacement designs, while the  
1094 complete-footprint results are more sensitive to the surrounding control prescription.